From Prime Number Theorem to Vinogradov's Theorem

Anshul Raj Singh

Indian Statistical Institute, Bangalore

anshulrsing@gmail.com

Supervisor: Dr. Alessandro Fazzari, University of Genova, Italy

October 23, 2026

Abstract

This exposition presents an introduction to several central topics in analytic number theory, culminating in a proof of the ternary Goldbach conjecture. We begin by studying the distribution of prime numbers through a proof of the Prime Number Theorem with an explicit error term that is

$$\pi(x) = \mathrm{li}(x) + O\left(\frac{x}{\exp(c\sqrt{\log x})}\right)$$

for some suitable positive constant c.

We then turn to primes in arithmetic progressions, discussing Dirichlet's theorem on the infinitude of primes in arithmetic progressions, the Prime Number Theorem for Arithmetic Progressions, and the Siegel-Walfisz theorem.

Finally we introduce the Hardy-Littlewood circle method and apply it to prove Vinogradov's theorem, from which the ternary Goldbach conjecture follows.

The exposition assumes a background in real and complex analysis together with a reasonable degree of mathematical maturity.

Acknowledgements

I would like to thank Dr. Alessandro Fazzari, for all his advice and guidance throughout the preparation of this exposition and for clearing up my doubts whenever I was stuck.

Contents

1 Preliminaries **5**
- 1.1 Dirichlet Series . . . 6
  - 1.1.1 Definitions and Elementary Results . . . 6
  - 1.1.2 Analytic Properties of Dirichlet series . . . 7
- 1.2 Euler Product and Zeta Function . . . 8
- 1.3 Perron's Formula . . . 12

2 The Prime Number Theorem **16**
- 2.1 The Setup . . . 16
- 2.2 A zero-free region for ζ . . . 16
  - 2.2.1 Auxiliary lemmas from complex analysis . . . 16
  - 2.2.2 Partial fraction decomposition of f'/f . . . 18
  - 2.2.3 Proof of the Zero-Free region . . . 20
- 2.3 Bounds on ζ'/ζ in the Zero-Free region . . . 21
- 2.4 The Prime Number Theorem . . . 22

3 Primes in Arithmetic Progression **25**
- 3.1 Dirichlet Characters . . . 25
- 3.2 Dirichlet L-functions . . . 26
- 3.3 PNT in Arithmetic Progression . . . 29
  - 3.3.1 Approximate Partial Fraction Decomposition of L'/L . . . 30
- 3.4 A Zero-Free Region for Dirichlet's L Functions. . . . 31
- 3.5 Bounds on L'/L in the Zero-Free Region . . . 34
- 3.6 Proof of PNT in Arithmetic Progressions . . . 35
- 3.7 Siegel-Walfisz theorem . . . 38

4 The Ternary Goldbach Conjecture **39**
- 4.1 Introduction . . . 39
  - 4.1.1 Major and Minor Arcs . . . 41
- 4.2 Auxiliary lemmas . . . 42
- 4.3 Estimation of Sums of Prime Exponentials . . . 43
- 4.4 Proof of Theorem 1 . . . 48
  - 4.4.1 Contribution of Major Arcs . . . 48
  - 4.4.2 Contribution of Minor Arcs . . . 52

Chapter 1

Preliminaries

In 1798 Adrien-Marie Legendre, on the basis of numerical evidence, conjectured that the number of primes less than or equal to x, which is denoted by $\pi(x)$, follows the asymptotic formula

$$\pi(x) \sim \frac{x}{\log x}.$$

This conjecture was later refined by Gauss, who suggested a more precise approximation using the logarithmic integral function. Numerical studies led him to observe that the density of primes around a large number x is approximately $1/\log x$, and thus the number of primes less than or equal to x can be estimated by integrating this density function, leading to the logarithmic integral

$$\mathrm{li}(x) = \int_2^x \frac{dt}{\log t}.$$

Integrating by parts we see that

$$\begin{aligned}
\mathrm{li}(x) &= \frac{x}{\log x} - \frac{2}{\log 2} + \int_2^x \frac{dt}{(\log t)^2} \\
&= \frac{x}{\log x} + O(1) + \int_2^{\sqrt{x}} \frac{dt}{(\log t)^2} + \int_{\sqrt{x}}^x \frac{dt}{(\log t)^2} \\
&= \frac{x}{\log x} + O(1) + O(\sqrt{x}) + O\left(\frac{x}{(\log x)^2}\right) \\
&= \frac{x}{\log x} + O\left(\frac{x}{(\log x)^2}\right).
\end{aligned}$$

Repeating this we see that for any positive integer K

$$\mathrm{li}(x) = x \sum_{k=1}^{K} \frac{(k-1)!}{(\log x)^k} + O_K\left(\frac{x}{(\log x)^K}\right).$$

From this one might conclude that the the error term in the approximation of $\pi(x)$ by $\mathrm{li}(x)$ has to be of the form $x/f(x)$ where $f(x) \gg (\log x)^K$ for any positive integer K.

In the next chapter we shall prove the Prime Number Theorem in the sharper quantitative form

$$\pi(x) = \mathrm{li}(x) + O\left(\frac{x}{\exp(c\sqrt{\log x})}\right)$$

for some suitable positive constant c. Note that $\exp(c\sqrt{\log x})$ grows much faster than any power of $\log x$, which is what we predicted f to be.

A common tool in analytic number theory for tackling such problems is the use of Dirichlet series which we shall define in the next section and study some of their properties.

1.1 Dirichlet Series

1.1.1 Definitions and Elementary Results

Definition 1.1. *A Dirichlet series is a series of the form*

$$\alpha(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

where s is a complex variable.

If we have another Dirichlet series $\beta(s) = \sum_{n=1}^{\infty} b_n n^{-s}$, then treating these sums as formal objects for now, we can define the product of these two Dirichlet series as

$$\alpha(s)\beta(s) = \sum_{n=1}^{\infty} \frac{c_n}{n^s},$$

where $c_n = \sum_{d|n} a_d b_{n/d}$, i.e. c_n is the Dirichlet convolution of the sequences (a_n) and (b_n). Treating (a_n) and (b_n) as arithmetic functions we see that

$$\alpha(s)\beta(s) = \sum_{n=1}^{\infty} \frac{(a * b)(n)}{n^s}.$$

Note: Throughout this exposition, we follow the convention that $s = \sigma + it$ and $s_k = \sigma_k + it_k$ for $\sigma, \sigma_k, t, t_k \in \mathbb{R}$ and a any positive integer $k \geq 0$.

A famous example of the Dirichlet series is the Riemann zeta function, which is defined by the absolutely convergent Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

for $\sigma > 1$. If we let $\alpha(s) = \beta(s) = \zeta(s)$, then

$$\zeta(s)^2 = \sum_{n=1}^{\infty} \frac{\mathbf{1} * \mathbf{1}(n)}{n^s} = \sum_{n=1}^{\infty} \frac{d(n)}{n^s},$$

where $d(n)$ denotes the number of divisors of n.

Using the zeta function Euler proved the infinitude of primes by showing that[1]

$$\zeta(s) = \prod_{p} \left(1 - \frac{1}{p^s}\right)^{-1},$$

[1] $\prod_p$ and $\sum_p$ indicate the product and sum over all primes, respectively.

and letting $s \to 1^+$ (approaching 1 from the right half-plane), and noting that the left hand side diverges implying that there cannot be finitely many primes.

He also showed that

$$\sum_p \frac{1}{p} = \infty,$$

which is a stronger result than the infinitude of primes. Euler's arguments were not rigorous by modern standards as complex analysis was not yet developed. To prove these results rigorously we study the properties of Dirichlet series in detail.

1.1.2 Analytic Properties of Dirichlet series

Theorem 1.2. *Suppose that the Dirichlet series $\alpha(s) = \sum_{n=1}^{\infty} a_n n^{-s}$ converges at the point $s = s_0$, and that $H > 0$ is an arbitrary constant. Then the series $\alpha(s)$ is uniformly convergent in the sector $\mathcal{S} = \{s \in \mathbb{C} : \sigma \geq \sigma_0, |t - t_0| \leq H(\sigma - \sigma_0)\}$.*

Proof. Let $R(u) = \sum_{n>u} a_n n^{-s_0}$ be the remainder of the series $\alpha(s_0)$. Using the Riemann-Stieljes integral we have

$$\sum_{n=M+1}^{N} a_n n^{-s} = -\int_M^N u^{s_0-s} dR(u) = -u^{s_0-s}R(u)\Big|_M^N + \int_M^N R(u) du^{s_0-s}$$

$$= M^{s_0-s}R(M) - N^{s_0-s}R(N) + (s_0 - s)\int_M^N R(u)u^{s_0-s-1}du. \quad (1.1)$$

If $|R(u)| \leq \varepsilon$ for all $u \geq M$ and if $\sigma > \sigma_0$, then from above we see that

$$\left|\sum_{n=M+1}^{N} \frac{a_n}{n^s}\right| \leq 2\varepsilon + \varepsilon|s - s_0| \int_M^{\infty} u^{\sigma_0-\sigma-1}du \leq \left(2 + \frac{|s - s_0|}{\sigma - \sigma_0}\right)\varepsilon.$$

But in the $\mathcal{S}$, $|s - s_0|/(\sigma - \sigma_0)$ is bounded, so the result follows by Cauchy's criterion. □

Corollary 1.3. *Any Dirichlet series $\alpha(s)$ has an abscissa of convergence σ_c such that $\alpha(s)$ converges for all s with $\sigma > \sigma_c$ and for no s with $\sigma < \sigma_c$. Moreover, if s_0 is a point with $\sigma_0 > \sigma_c$, then there is a neighbourhood of s_0 in which $\alpha(s)$ converges uniformly.*

It follows that the series $\alpha(s) = \sum a_n n^{-s}$ is analytic for $\sigma > \sigma_c$, and that the differentiated series is locally uniformly convergent to $\alpha'(s)$

$$\alpha'(s) = -\sum_{n=1}^{\infty} \frac{a_n}{n^s} \log n$$

for s in the half-plane $\sigma > \sigma_c$.

Remark 1.4. *If s_0 is at the line of convergence (i.e., $\sigma_0 = \sigma_c$), and that the series $\alpha(s_0)$ converges. Then $\lim_{s\to s_0} \alpha(s)$ with $\sigma > \sigma_c$ need not exist. However $\lim_{s\to s_0} \alpha(s) = \alpha(s_0)$ for $s \in \mathcal{S}$.*

Theorem 1.5. *Let $A(x) = \sum_{n \le x} a_n$. If $\sigma_c < 0$, then $A(x)$ is a bounded function, and*

$$\sum_{n=1}^{\infty} a_n n^{-s} = s \int_1^{\infty} A(x) x^{-s-1} dx \tag{1.2}$$

for $\sigma > 0$. If $\sigma_c \ge 0$, then

$$\limsup_{x \to \infty} \frac{\log |A(x)|}{\log x} = \sigma_c, \tag{1.3}$$

and (1.2) holds for $\sigma > \sigma_c$.

Proof. We note that

$$\begin{aligned} \sum_{n=1}^{N} a_n n^{-s} &= \int_{1^-}^{N} x^{-s} dA(x) = A(x) x^{-s} \Big|_{1^-}^{N} + \int_{1^-}^{N} A(x) dx^{-s} \\ &= A(N) N^{-s} + s \int_1^N A(x) x^{-s-1} dx. \end{aligned}$$

Let ϕ denote the left-hand side of (1.3). If $\theta > \phi$ then $A(x) \ll x^{\theta}$ where the implicit constant may depend on the a_n and on θ. Thus if $\sigma > \theta$, then the integral in (1.2) is absolutely convergent. Thus by letting $N \to \infty$, we obtain the desired result, we obtain (1.2). It is clear that $A(x)$ is bounded for $\sigma_c < 0$, and that (1.2) holds.

Now suppose that $\sigma_c \ge 0$. By corollary 1.3 the series in (1.2) diverges for $\sigma < \sigma_c$. Hence $\phi \ge \sigma_c$. Now choose $\sigma_0 > \sigma_c$. By (1.1) with $s = 0$ and $M = 0$, we see that

$$A(N) = -R(N) N^{\sigma_0} + \sigma_0 \int_0^N R(u) u^{\sigma_0 - 1} du.$$

Since $R(u)$ is bounded function, it follows that $A(N) \ll N^{\sigma_0}$ where the implicit constant may depend on the σ_0. Hence $\phi \le \sigma_0$. Since this holds for any $\sigma_0 > \sigma_c$, we conclude that $\phi \le \sigma_c$. □

The following result follows which we leave as an exercise to the reader.

Theorem 1.6. *With the general notation, $\sigma_c \le \sigma_a \le \sigma_c + 1$.*

1.2 Euler Product and Zeta Function

Formal calculations with Dirichlet series are justified once they are absolutely convergent. Euler factorization being one of them. We study the Euler products in this section.

We leave the proof of the following theorem as an exercise to the reader.

Theorem 1.7. *Let $\alpha(s) = \sum a_n n^{-s}$ and $\beta(s) = \sum b_n n^{-s}$ be two Dirichlet series, and put $\gamma(s) = \sum c_n n^{-s}$ where $c_n = (a * b)(n)$. If $\alpha(s)$ and $\beta(s)$ are absolutely convergent at some point s, then $\gamma(s)$ is absolutely convergent and $\gamma(s) = \alpha(s)\beta(s)$.*

We now come to the factorization of Dirichlet functions. An arithmetic function $f(n)$ is said to be **multiplicative** if $f(1) = 1$ and $f(mn) = f(m)f(n)$ for all m, n with $(m, n) = 1$. It is called **totally multiplicative** if $f(1) = 1$ and $f(mn) = f(m)f(n)$ for all m, n. If f is multiplicative then the Dirichlet series $\sum f(n)n^{-s}$ factors into product over primes. We first give a formal argument to see this. Note that when the product

$$\prod_p (1 + f(p)p^{-s} + f(p^2)p^{-2s} + f(p^3)p^{-3s} + \cdots) = \prod_p \left(1 - f(p)p^{-s}\right)^{-1}$$

is expanded, the generic term looks like

$$\frac{f(p_1^{k_1})f(p_2^{k_2})\cdots f(p_r^{k_r})}{(p_1^{k_1}p_2^{k_2}\cdots p_r^{k_r})^s}.$$

Therefore, by the fundamental theorem of arithmetic and multiplicativity of f, and rearranging, we see that this product is equal to the Dirichlet sum.

Theorem 1.8. *If f is multiplicative and $\sum |f(n)|n^{-\sigma} < \infty$, then*

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_p \left(1 - f(p)p^{-s}\right)^{-1}.$$

Proof. For any prime p,

$$\sum_{k=0}^{\infty} |f(p^k)|p^{-k\sigma} \leq \sum_{n=1}^{\infty} |f(n)|n^{-\sigma} < \infty,$$

so we can use the sum of geometric series to get

$$(1 + f(p)p^{-s} + f(p^2)p^{-2s} + f(p^3)p^{-3s} + \cdots) = \left(1 - f(p)p^{-s}\right)^{-1}$$

for any prime p. Let y be a positive real number, and let $1 \in \mathcal{N}$ be the set of those positive integers composed entirely of primes not exceeding y. Since a product of finitely many absolutely convergent series may be arbitrarily rearranged, we see that

$$\Pi_y = \prod_{p \leq y} \left(1 + f(p)p^{-s} + f(p^2)p^{-2s} + \cdots\right) = \sum_{n \in \mathcal{N}} f(n)n^{-s}.$$

Hence

$$\left|\Pi_y - \sum_{n=1}^{\infty} f(n)n^{-s}\right| \leq \sum_{n \notin \mathcal{N}} |f(n)|n^{-\sigma} \leq \sum_{n > y} |f(n)|n^{-\sigma}. \tag{1.4}$$

Letting $y \to \infty$ gives the result. □

As a corollary we get

Corollary 1.9. *For $\sigma > 1$,*

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s) = \prod_p \left(1 - p^{-s}\right)^{-1}.$$

Therefore, letting $s \to 1^+$ we see that there are infinitely many primes. Taking logarithms on both sides with $\sigma > 1$, we get

$$-\sum_p \log(1 - p^{-s}) = \log \zeta(s).$$

Since $\log(1+z) = z + O(|z|^2)$ for $|z| \leq 1/2$, we get

$$-\sum_p \left(-\frac{1}{p^s} + O\left(\frac{1}{p^{2s}}\right)\right) = \log \zeta(s)$$
$$\implies \sum_p \frac{1}{p^s} + O(1) = \log \zeta(s).$$

The error term is $O(1)$ because the sum converges. Again letting $s \to 1$ with $\sigma > 1$, we get that the sum of reciprocals of primes diverges. Infact the following is true

Proposition 1.10. *For $N \in \mathbb{N}$ we have*

$$\sum_{p<N} \frac{1}{p} = \log\log N + O(1).$$

Before proving this result we collect some important expressions and estimates for zeta function.

Theorem 1.11. *Suppose that $\sigma > 0, x > 0$ and that $s \neq 1$. Then*

$$\zeta(s) = \sum_{n \leq x} \frac{1}{n^s} + \frac{x^{1-s}}{s-1} + \frac{\{x\}}{x^s} - s\int_x^\infty \{u\}u^{-s-1}du.$$

In particular, for $x = 1$ we get

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{u\}u^{-s-1}du \tag{1.5}$$

for $\sigma > 0$.

Proof. For $\sigma > 1$ we have

$$\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s} = \sum_{n \leq x} \frac{1}{n^s} + \sum_{n > x} \frac{1}{n^s}.$$

We write the second sum as

$$\int_x^\infty u^{-s} d[u] = \int_x^\infty u^{-s}du - \int_x^\infty u^{-s}d\{u\}.$$

We evaluate the first integral on the right-hand side, and integrate the second one by parts. Thus te above is

$$= \frac{x^{1-s}}{s-1} + \{x\}x^{-s} + \int_x^\infty \{u\}du^{-s} = \frac{x^{1-s}}{s-1} + \{x\}x^{-s} - s\int_x^\infty \{u\}u^{-s-1}du.$$

This completes the proof. □

Here are some important corollaries that follow.

Corollary 1.12. *The Riemann zeta function has a simple pole at $s = 1$ with residue 1, but is otherwise analytic in the half-plane $\sigma > 0$.*

Corollary 1.13. *The inequalities*

$$\frac{1}{\sigma - 1} < \zeta(\sigma) < \frac{\sigma}{\sigma - 1}$$

hold for all $\sigma > 0$. In particular, $\zeta < 0$ for $0 < \sigma < 1$.

Corollary 1.14. *Let $\delta > 0$ be fixed. Then for $\sigma \geq \delta$, $s \neq 1$,*

$$\sum_{n \leq x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(\tau x^{-\sigma}).$$

In addition,

$$\sum_{n \leq x} \frac{1}{n} = \log x + C_0 + O(x^{-1})$$

where C_0 is Euler's constant,

$$C_0 = 1 - \int_1^\infty \{u\} u^{-2} du = 0.5772156649\ldots.$$

Proof. Left as an exercise to the reader. □

Corollary 1.15. *Let*

$$\zeta(s) = \frac{1}{s-1} + \sum_{k=0}^{\infty} a_k (s-1)^k$$

be the Laurent expansion of $\zeta(s)$ as $s = 1$. Then a_0 is the Euler's constant.

Corollary 1.16. *Let $\delta > 0$ be fixed. Then*

$$\zeta(s) = \frac{1}{s-1} + O(1)$$

uniformly for s in the rectangle $\delta \leq \sigma \leq 2$, $|t| \leq 1$, and

$$\zeta(s) \ll (1 + \tau^{1-\sigma}) \min\left(\frac{1}{|\sigma - 1|}, \log \tau\right)$$

uniformly for $\delta \leq \sigma \leq 2$, $|t| \geq 1$.

Proof. The first assertion is clear from (1.5). When $|t|$ is larger, we obtain a bound for $|\zeta(s)|$ by estimating the sum in corollary 1.14. Assume that $x \geq 2$. We observe that

$$\sum_{n \leq x} n^{-\sigma} \ll \sum_{1 \leq n \leq x} n^{-\sigma} < 1 + \int_1^x u^{-\sigma}\, du$$

uniformly for $\sigma \geq 0$. If $0 \leq \sigma \leq 1 - 1/\log x$, then this integral is

$$\frac{x^{1-\sigma} - 1}{1-\sigma} < \frac{x^{1-\sigma}}{1-\sigma}.$$

If $|\sigma - 1| \leq 1/\log x$, then $u^{-\sigma} \ll u^{-1}$ uniformly for $1 \leq u \leq x$, and hence the integral is

$$\ll \int_1^x u^{-1}\, du = \log x.$$

If $\sigma \geq 1 + 1/\log x$, then the integral is

$$\int_1^\infty u^{-\sigma}\, du = \frac{1}{\sigma - 1}.$$

Thus

$$\sum_{n \leq x} n^{-s} \ll (1 + x^{1-\sigma}) \min\left(\frac{1}{|\sigma - 1|}, \log x\right) \tag{1.6}$$

uniformly for $0 \leq \sigma \leq 2$. The second assertion now follows by taking $x = \tau$ in corollary 1.14. □

Proof of the proposition 1.10. Let $\sigma > 1$. From the proof of theorem 1.8, we get that for a positive (large) real number y

$$\left|\prod_{p \leq y} \left(1 - p^{-\sigma}\right)^{-1} - \sum_{n \leq y} \frac{1}{n^\sigma}\right| = O(1) \implies \prod_{p \leq y} \left(1 - p^{-\sigma}\right)^{-1} = \sum_{n \leq y} \frac{1}{n^\sigma} + O(1)$$

where the implicit constant depends on y. Therefore, taking logarithms we get

$$\sum_{p \leq y} \frac{1}{p^\sigma} = \log \sum_{n \leq y} \frac{1}{n^\sigma} + O(1).$$

Letting $\sigma \to 1^+$ we get the result. □

Inspired by Euler's methods Dirichlet (1837) proved that there are infinitely many primes in an arithmetic progression $\{a + nq\}_{n \in \mathbb{N}}$ where $a, q \in \mathbb{N}$ and $(a, q) = 1$. He proved that the sum

$$\sum_{p \equiv a \pmod q} \frac{1}{p} = \infty.$$

We will see a proof of this theorem in chapter 3.

1.3 Perron's Formula

We now introduce a powerful tool to deal with sums like $A(x)$.

Lemma 1.17. *If $\sigma_0 > 0$, then*

$$\frac{1}{2\pi i}\int_{\sigma_0-i\infty}^{\sigma_0+i\infty} y^s \frac{ds}{s} = \begin{cases} 1 & \text{if } y > 1, \\ 1/2 & \text{if } y = 1, \\ 0 & \text{if } 0 < y < 1. \end{cases}$$

Proof. Left as an exercise. □

Theorem 1.18 (Perron's Formula)**.** *If $\sigma_0 > \max(0, \sigma_c)$, and $x > 0$, then*

$$\sum_{n\leq x}{}' a_n = \lim_{T\to\infty} \frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT} \alpha(s)\frac{x^s}{s}ds.$$

Here $\sum'$ indicates that if x is an integer, then the last term is to be counted with weight $1/2$.

Proof. Choose N so large that $N > 2x + 2$, and write

$$\alpha(s) = \sum_{n\leq N} a_n n^{-s} + \sum_{n>N} a_n n^{-s} = \alpha_1(s) + \alpha_2(s).$$

Note that

$$\sum_{n\leq x}{}' a_n = \lim_{T\to\infty} \frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT} \alpha_1(s)\frac{x^s}{s}ds.$$

For $\alpha_2(s)$, Riemann-Stieljes implies

$$\alpha_2(s) = \int_N^\infty u^{-s} d(A(u) - A(N)) = s\int_N^\infty (A(u) - A(N))u^{-s-1}du.$$

But $A(u) - A(N) \ll u^\theta$ for $\theta > \max(0, \sigma_c)$, and hence

$$\alpha_2(s) \ll \left(1 + \frac{|s|}{\sigma - \theta}\right) N^{\theta-\sigma}$$

for $\sigma > \theta > \max(0, \sigma_c)$. Implicit constants here and in the rest of this proof may depend on the a_n. Hence

$$\int_{\sigma_0\pm iT}^{T\pm iT} \alpha_2(s)\frac{x^s}{s}ds \ll \frac{N^\theta}{\sigma_0 - \theta}\int_{\sigma_0}^\infty \left(\frac{x}{N}\right)^\sigma d\sigma \ll \frac{N^\theta}{\sigma_0 - \theta}\frac{(x/N)^{\sigma_0}}{\log N/x},$$

and

$$\int_{T-iT}^{T+iT} \alpha_2(s)\frac{x^s}{s}ds \ll N^\theta (x/N)^{\sigma_0}$$

for large T. Take θ with $\sigma_0 > \theta > \max(0, \sigma_c)$. Hence by Cahuchy's criterion

$$\int_{\sigma_0-iT}^{\sigma_0+iT} = \int_{\sigma_0-iT}^{T-iT} + \int_{T-iT}^{T+iT} + \int_{\sigma_0+iT}^{T+iT} \ll x^{\sigma_0} N^{\theta-\sigma_0}.$$

On combining our estimates, we see that

$$\limsup_{T\to\infty}\left|\sideset{}{'}\sum_{n\leq x}a_n-\frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT}\alpha(s)\frac{x^s}{s}ds\right|\ll x_0^{\sigma}N^{\theta-\sigma_0}.$$

Letting $N\to\infty$, this lim sup is 0, and the proof is complete. □

The formula itself is not very useful in practice because of the sums involved. Instead we look at the truncated Perron's formula and express the error term in terms of the sine integral

$$\mathrm{si}(x)=-\int_x^{\infty}\frac{\sin u}{u}du.$$

By integratino by parts we see that $\mathrm{si}(x)\ll 1/x$ for $x\geq 1$, and hence that

$$\mathrm{si}(x)\ll\min(1,1/x) \tag{1.7}$$

for $x>0$. We also note that

$$\mathrm{si}(x)+\mathrm{si}(-x)=-\int_{-\infty}^{+\infty}\frac{\sin u}{u}du=-\pi. \tag{1.8}$$

Theorem 1.19. *If $\sigma_0>\max(0,\sigma_a)$ and $x>0$, then*

$$\sideset{}{'}\sum_{n\leq x}a_n=\frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT}\alpha(s)\frac{x^s}{s}ds+R$$

where

$$R=\frac{1}{\pi}\sum_{x/2<n<2x}a_n si\left(T\log\frac{x}{n}\right)-\frac{1}{\pi}\sum_{x<n<2x}a_n si\left(T\log\frac{n}{x}\right)+O\left(\frac{x^{\sigma_0}}{T}\sum_n\frac{|a_n|}{n^{\sigma_0}}\right).$$

Proof. Since the series $\alpha(s)$ is absolutely convergent on the interval $[\sigma_0-iT,\sigma_0+iT]$, we see that

$$\frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT}\alpha(s)\frac{x^s}{s}ds=\sum_n a_n\frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT}\left(\frac{x}{n}\right)^s\frac{ds}{s}.$$

It suffices to show that

$$\frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT}y^s\frac{ds}{s}=\begin{cases}1+O(y^{\sigma_0}/T) & \text{if } y\geq 2,\\ 1+\frac{1}{\pi}\mathrm{si}(T\log y)+O(2^{\sigma_0}/T) & \text{if } 1\leq y\leq 2,\\ -\frac{1}{\pi}\mathrm{si}(T\log 1/y)+O(2^{\sigma_0}/T) & \text{if } 1/2\leq y\leq 1,\\ O(y^{\sigma_0}/T) & \text{if } y\leq 1/2\end{cases} \tag{1.9}$$

for $\sigma_0>0$.

For the first part, suppose that $y\geq 2$, and let $\mathcal{C}$ be the piecewise linear path from $-\infty-iT$ to σ_0-iT to σ_0+iT to $-\infty+iT$. Then by the residue theorem we see that

$$\frac{1}{2\pi i}\int_{\mathcal{C}}y^s\frac{ds}{s}=1.$$

In addition,

$$\int_{-\infty\pm iT}^{\sigma_0\pm iT} y^s\frac{ds}{s} = \int_{-\infty}^{\sigma_0}\frac{y^{\sigma\pm iT}}{\sigma\pm iT}d\sigma \ll \frac{1}{T}\int_{-\infty}^{\sigma_0} y^\sigma d\sigma = \frac{y^{\sigma_0}}{T\log y} \ll \frac{y^{\sigma_0}}{T},$$

so we have (1.9) for $y \geq 2$. The case $y \leq 1/2$ is treated similarly with the contour shifted to the right, and then there is no residue.

Suppose now that $1 \leq y \leq 2$, and take $\mathcal{C}$ to be the closed rectangular path from $\sigma_0 - iT$ to $\sigma_0 + iT$ to iT to $-iT$ to $\sigma_0 - iT$, with a semicircular indentation of radius ε at $s = 0$. Then by Cauchy's theorem

$$\frac{1}{2\pi i}\int_{\mathcal{C}} y^s\frac{ds}{s} = 0.$$

We note that

$$\int_{\pm iT}^{\sigma_0\pm iT} y^s\frac{ds}{s} \ll \frac{1}{T}\int_0^{\sigma_0} y^\sigma d\sigma \ll \frac{s^{\sigma_0}}{T}.$$

The integral around the semicircle tends to $1/2$ as $\varepsilon \to 0$, and the remaining intehral is

$$\begin{aligned}\frac{1}{2\pi i}\lim_{\varepsilon\to 0}\left(\int_{i\varepsilon}^{iT} + \int_{-iT}^{-i\varepsilon}\right) y^s\frac{ds}{s} &= \frac{1}{2\pi i}\lim_{\varepsilon\to 0}\int_\varepsilon^T \left(y^{it} - y^{-it}\right)\frac{dt}{t}\\ &= \frac{1}{\pi}\int_0^{T\log y}\sin v\frac{dv}{v}\\ &= \frac{1}{2} + \frac{1}{\pi}\mathrm{si}\left(T\log y\right)\end{aligned}$$

by (1.8). This gives (1.9) when $1 \leq y \leq 2$ and the case $1/2 \leq y \leq 1$ is treated similarly. □

The error term is still complicated and is not much useful in calculations. Instead we use the following which is more convenient.

Corollary 1.20. *In theorem (1.19),*

$$R \ll \sum_{\substack{x/2<n<2x\\ n\neq x}} |a_n|\min\left(1, \frac{x}{T|x-n|}\right) + \frac{x^{\sigma_0}}{T}\sum_{n=1}^{\infty}\frac{|a_n|}{n^{\sigma_0}}.$$

Proof. From (1.7) we see that

$$\mathrm{si}\left(T|\log n/x|\right) \ll \min\left(1, \frac{1}{T|\log n/x|}\right).$$

But $n/x = 1 + (n-x)/x$ and $|\log(1+\delta)| \asymp |\delta|$ uniformly for $-1/2 \leq \delta \leq 1$, so the above is

$$\asymp \min\left(1, \frac{x}{T|x-n|}\right)$$

if $x/2 \leq n \leq 2x$. By theorem 1.19 the result follows. □

Chapter 2

The Prime Number Theorem

2.1 The Setup

By the Perron's formula, for $\sigma_0 > 1$, and $x > 0$, we have

$$\psi(x) := \sum_{n \leq x} \Lambda(n) = -\frac{1}{2\pi i} \int_{\sigma_0 - iT}^{\sigma_0 + iT} \frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds + R, \tag{2.1}$$

we know R explicitly. For the first term, the integrand has a simple pole at the point $s = 1$, so we would like to apply residue theorem in some region containing this pole and hence we would like to understand the behaviour of $\zeta'(s)/\zeta(s)$. We prove that $\zeta(s) \neq 0$ for $\sigma > 1 - \eta(t)$ for some decreasing function η.

2.2 A zero-free region for ζ

Theorem 2.1. *There is an absolute constant $c > 0$ such that $\zeta(s) \neq 0$ for $\sigma \geq 1 - c/\log \tau$, where $\tau = |t| + 4$.*

2.2.1 Auxiliary lemmas from complex analysis

Lemma 2.2. *(Jensen's Inequality) Let f is analytic in a domain containing the disc $|z| \leq R$. Suppose f satisfies $|f(z)| \leq M$ in this disc and that $f(0) \neq 0$. Then, for $r < R$, the number of zeroes of f in the disc $|z| \leq r$ does not exceed*

$$\frac{\log M/|f(0)|}{\log R/r}.$$

Proof. First note that f cannot have infinitely many zeros in the compact disc $|z| \leq R$ otherwise we would have a converging sequence of zeros of f which is impossible since f is non-constant (we are using the identity theorem here). Now let $z_1, z_2, \ldots, z_K$ denote the zeros of f in the disc $|z| \leq R$ counted with multiplicities, and set

$$g(z) = f(z) \prod_{k=1}^{K} \frac{R^2 - z\overline{z_k}}{R(z - z_k)}.$$

Note that since g is bounded near each z_i, by Reimann's theorem, z_i's are removable singularities of g, so we can assume without loss of generality that g is analytic in the disc $|z| \leq R$. Also, note that if $|z| = R$, then $|g(z)| = |f(z)| \leq M$. Hence by the maximum modulus principle, $|g(z)| \leq M$, in particular, $|g(0)| \leq M$. But

$$M \geq |g(0)| = |f(0)| \prod_{k=1}^{K} \frac{R}{|z_k|}.$$

Each factor in the product is at least 1, and if there are $\mathbf{n}(r)$ zeros inside the disc $|z| \leq r$, then

$$\frac{M}{|f(0)|} \geq \left(\frac{R}{r}\right)^{\mathbf{n}(r)} \Rightarrow \mathbf{n}(r) \leq \frac{\log M/|f(0)|}{\log R/r}.$$

This completes the proof. □

Remark 2.3. *A second proof of this is by using Jensen's formula:*

$$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta,$$

where the z_k are the zeros of f inside the disc D_R (counted with multiplicities), f is holomorphic in an open set Ω containing the closure of the disc D_R with $f(0) \neq 0$, and f vanishes nowhere on the boundary of D_R.
Using this, we get:

$$\log M - \log|f(0)| \geq \sum_{k=1}^{K} \log\left(\frac{R}{|z_k|}\right) \geq \boldsymbol{n}(r) \log\left(\frac{R}{r}\right) \implies \boldsymbol{n}(r) \leq \frac{\log M/|f(0)|}{\log R/r}.$$

Lemma 2.4. *(The Borel-Carathéodory Lemma) Suppose that $h(z)$ is analytic in a domain containing the disc $|z| \leq R$, with $h(0) = 0$ and $\Re h(z) \leq M$ in this disc. If $|z| < r < R$, then*

$$|h(z)| \leq \frac{2Mr}{R-r},$$

and

$$|h'(z)| \leq \frac{2MR}{(R-r)^2}.$$

Proof. We prove the following general result

$$\left|\frac{h^{(k)}(0)}{k!}\right| \leq \frac{2M}{R^k} \tag{2.2}$$

for all $k \geq 1$. This with the power series expansion and triangle inequality implies

$$|h(z)| \leq \sum_{k=1}^{\infty} \left|\frac{h^{(k)}(0)}{k!}\right| r^k \leq 2M \sum_{k=0}^{\infty} \left(\frac{r}{R}\right)^k \leq \frac{2Mr}{(R-r)},$$

and

$$|h'(z)| \leq \sum_{k=1}^{\infty} \left|\frac{h^{(k)}(0)}{k!}\right| kr^{k-1} \leq \frac{2M}{R} \sum_{k=1}^{\infty} k \left(\frac{r}{R}\right)^{k-1} = \frac{2MR}{(R-r)^2}.$$

To prove (2.2) first note that by the Cauchy's integral formula, we have

$$\int_0^1 h(Re^{2\pi i\theta})d\theta = \frac{1}{2\pi i} \oint_{|z|=R} \frac{h(z)}{z} dz = h(0) = 0.$$

Moreover, if $k \geq 1$,

$$\int_0^1 h(Re^{2\pi i\theta})e^{2k\pi i\theta} d\theta = \frac{R^{-k}}{2\pi i} \oint_{|z|=R} h(z)z^{k-1} dz = 0,$$

and

$$\int_0^1 h(Re^{2\pi i\theta})e^{-2k\pi i\theta} d\theta = \frac{R^k}{2\pi i} \oint_{|z|=R} h(z)z^{-k-1} dz = \frac{R^k h^{(k)}(0)}{k!}.$$

Taking an appropriate linear combination of these equations, for $k > 0$, we have

$$\int_0^1 h(Re^{2\pi i\theta})(1 + \cos 2\pi(k\theta + \phi))d\theta = \frac{R^k e^{-2\pi i\phi} h^{(k)}(0)}{2k!}.$$

By taking real parts it follows that

$$\begin{aligned}
\Re\left(\frac{R^k e^{-2\pi i\phi} h^{(k)}(0)}{2k!}\right) &= \Re\left(\int_0^1 h(Re^{2\pi i\theta})\big(1 + \cos(2\pi(k\theta + \phi))\big)\, d\theta\right) \\
&\leq M \int_0^1 \big(1 + \cos(2\pi(k\theta + \phi))\big)\, d\theta \\
&= M.
\end{aligned}$$

Since the inequality holds for all ϕ, we choose ϕ so that $|h^{(k)}(0)| = e^{-2\pi i\phi} h^{(k)}(0)$, proving (2.2). □

2.2.2 Partial fraction decomposition of f'/f

Lemma 2.5. *Suppose that f is analytic in a domain containing the disc $|z| \leq 1$, with $|f(z)| \leq M$ in the disc and $f(0) \neq 0$. Let $0 < r < R < 1$ be fixed. Then for $|z| \leq r$ we have*

$$\frac{f'}{f}(z) = \sum_{k=1}^{K} \frac{1}{z - z_k} + O\left(\log \frac{M}{|f(0)|}\right),$$

where the sum is extended over all zeros z_k of f with $|z_k| \leq R$. (The implicit constant depends on r and R, but is otherwise absolute.)

Proof. Let $z_1, \dots, z_K$ denote the zeros of f with $|z_k| < R$, counted with multiplicity, and assume that f is non-zero on $|z| = R$. Define

$$g(z) = f(z) \prod_{k=1}^{K} \frac{R^2 - z\overline{z_k}}{R(z - z_k)}.$$

The kth factor has a pole at z_k and is non-zero for $|z| \le R$, because $|z\overline{z_k}| \le R|z_k| < R^2$ when $|z| \le R$ and $|z_k| < R$. Hence g extends to an analytic function in $|z| \le R$. On $|z| = R$ each factor has modulus 1, so $|g(z)| = |f(z)| \le M$ there. By the maximum modulus principle, $|g(z)| \le M$ for $|z| \le R$.

Since $g(0) \neq 0$, the function

$$h(z) = \log \frac{g(z)}{g(0)}$$

is analytic in $|z| \le R$ and satisfies $\Re h(z) \le \log(M/|g(0)|)$. Applying the Borel–Carathéodory lemma to h with radius R gives

$$|h'(z)| \le \frac{2R}{(R-r)^2} \log \frac{M}{|g(0)|} \qquad (|z| \le r).$$

Since $g'/g = h'$, we obtain for $|z| \le r$,

$$\frac{f'}{f}(z) = \frac{g'}{g}(z) + \sum_{k=1}^{K} \frac{1}{z - z_k} = \sum_{k=1}^{K} \frac{1}{z - z_k} + O\left(\log \frac{M}{|g(0)|}\right).$$

Finally, $|g(0)| = |f(0)| \prod_{k=1}^{K} R/|z_k| \ge |f(0)|$, so the implied constant is absolute and the error term is $O(\log(M/|f(0)|))$. □

Corollary 2.6. *If $|t| \ge 7/8$ and $5/6 \le \sigma \le 2$, then*

$$\frac{\zeta'}{\zeta}(s) = \sum_{\rho} \frac{1}{s - \rho} + O(\log \tau)$$

where the sum is extended over all zeros ρ of $\zeta(s)$ for which $|\rho - (3/2 + it)| \le 5/6$.

Proof. Consider the function $f(z) = \zeta(z + 3/2 + it)$. In lemma 2.5 take $R = 5/6$ and $r = 2/3$. Since the pole of ζ occurs at 1, the only possible singularity of $f(z) = \zeta(z + 3/2 + it)$ is at $z = -1/2 - it$. For $|t| \ge 7/8$ this point lies outside the unit disc because $|-1/2 - it| \ge \sqrt{65}/8 > 1$. Hence f is analytic in a neighborhood of $|z| \le 1$.

Now we have a region in which f is analytic and it contains the unit disc $|z| \le 1$. Therefore, by lemma 2.5 we have

$$\frac{\zeta'}{\zeta}(z + 3/2 + it) = \sum_{\substack{\eta:\text{zero of } f \\ |\eta| \le 5/6}} \frac{1}{z - \eta} + O\left(\log \frac{M}{|f(0)|}\right).$$

Letting $s = z + 3/2 + it$ with $|z| \leq 2/3$ and $\rho = \eta + 3/2 + it$, we get

$$\frac{\zeta'}{\zeta}(s) = \sum_{\substack{\rho:\text{zero of } \zeta \\ |\rho-(3/2+it)|\leq 5/6}} \frac{1}{s-\rho} + O\left(\log \frac{M}{|f(0)|}\right).$$

To get the error term we know that for $\sigma > 0$, we have

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{u\}u^{-s-1}\,du.$$

Therefore, in the region $5/6 \leq \Re(s) \leq 13/6$ and $|t| \geq 7/8$ we have

$$\begin{aligned}|\zeta(s)| &\leq \left|\zeta(s) - \frac{s}{s-1}\right| + \left|\frac{s}{s-1}\right| \\ &\leq \left|\frac{s}{s-1}\right| + \frac{|s|}{\sigma} \ll |s| \ll \tau.\end{aligned}$$

Therefore, $M \ll \tau$, while $\zeta(3/2 + it) \gg 1$ by the (absolutely convergent) Euler product formula, giving us $\log(M/|f(0)|) \ll \log \tau$. Finally, note that $|z| \leq 2/3$ gives $5/6 \leq \sigma \leq 13/6$, which clearly contains the region $|t| \geq 7/8$ and $5/6 \leq \sigma \leq 2$. □

2.2.3 Proof of the Zero-Free region

We require one further lemma.

Lemma 2.7 (de la Vallée Poussin)**.** *If $\sigma > 1$, then*

$$\Re\left(-3\frac{\zeta'}{\zeta}(\sigma) - 4\frac{\zeta'}{\zeta}(\sigma+it) - \frac{\zeta'}{\zeta}(\sigma+2it)\right) \geq 0.$$

Proof. Using

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=1}^\infty \frac{\Lambda(n)}{n^s},$$

we see that the right hand side of the inequality is:

$$\sum_{n=1}^\infty \frac{\Lambda(n)}{n^\sigma}(3 + 4\cos(t\log n) + \cos(2t\log n)).$$

It suffices to note that $3 + 4\cos\theta + \cos 2\theta = 2(1+\cos\theta)^2 \geq 0$ for all θ. □

Proof of theorem 2.1. For $\sigma > 1$, $\zeta(s)$ is non zero via the Euler product. Let $\sigma \leq 1$. For $\sigma > 0$, using (1.5), we obtain:

$$\left|\zeta(s) - \frac{s}{s-1}\right| \leq |s|\int_1^\infty u^{-\sigma-1}du = \frac{|s|}{\sigma} \tag{2.3}$$

for $\sigma > 0$. From this we see that $\zeta(s) \neq 0$ when $\sigma > |s-1|$, i.e., in the parabolic region $\sigma > (1+t^2)/2$, particularly in the rectangle $8/9 \leq \sigma \leq 1$.

The main idea now is to find an absolute constant $c > 0$ such that if there is a zero, say $\rho_0 = \beta_0 + i\gamma_0$ of ζ then $\beta_0 < 1 - c/\log|\gamma_0| + 4$. This will prove the theorem.

Let $\rho = \beta_0 + i\gamma_0$ be a zero of ζ. Since $\Re(\rho) > 0$ for all zeros ρ of $\zeta(s)$, we find that $\Re(1/(s-\rho)) > 0$ whenever $\sigma > 0$. Hence by corollary 2.6, with $s = 1 + \delta + i\gamma_0$, we see that

$$\begin{aligned} -\Re\left(\frac{\zeta'}{\zeta}(1+\delta+i\gamma_0)\right) &= -\Re\left(\sum_{\rho}\frac{1}{1+\delta+i\gamma_0-\rho}\right) + O(\log\tau) \\ &\leq -\frac{1}{1+\delta-\beta_0} + c_1\log\left(|\gamma_0|+4\right), \end{aligned}$$

where c_1 is an absolute positive constant from corollary 2.6.
Similarly, by letting $s = 1 + \delta + 2i\gamma_0$, we obtain:

$$-\Re\left(\frac{\zeta'}{\zeta}(1+\delta+2i\gamma_0)\right) \leq c_1\log\left(|\gamma_0|+4\right).$$

Finally we have

$$-\Re\left(\frac{\zeta'}{\zeta}(1+\delta)\right) = \frac{1}{\delta} + O(1).$$

On combining these in lemma 2.7 we get

$$\frac{3}{\delta} - \frac{4}{1+\delta-\beta_0} + c_2\log\left(|\gamma_0|+4\right) \geq 0.$$

Taking $\delta = 1/2c_2\log\left(|\gamma_0|+4\right)$ we get:

$$\begin{aligned} 7c_2\log\left(|\gamma_0|+4\right) \geq \frac{4}{1+\delta-\beta_0} &\Rightarrow 1-\beta_0 \geq \frac{1}{14c_2\log\left(|\gamma_0|+4\right)} \\ &\Rightarrow \beta_0 \leq 1 - \frac{1}{14c_2\log\left(|\gamma_0|+4\right)}, \end{aligned}$$

where the constan c_2 is absolute. This completes the proof. □

2.3 Bounds on ζ'/ζ in the Zero-Free region

Theorem 2.8. *Let c be the constant in theorem 2.1. If $\sigma > 1 - c/(2\log\tau)$ and $|t| \geq 7/8$, then $\zeta'(s)/\zeta(s) \ll \log\tau$ and, if $1 - c/(2\log\tau) < \sigma \leq 2$ and $|t| \leq 7/8$, then $\zeta'(s)/\zeta(s) = -1/(s-1) + O(1)$.*

Proof. If $\sigma > 1$, we see that

$$\left|\frac{\zeta'}{\zeta}(s)\right| \leq \sum_{n=1}^{\infty} \Lambda(n) n^{-\sigma} = -\frac{\zeta'}{\zeta}(\sigma) \ll \frac{1}{\sigma - 1}.$$

Hence the first inequality is obvious if $\sigma \geq 1 + 1/\log T$. Let $s_1 = 1 + 1/\log T + it$. In particular we have

$$\frac{\zeta'}{\zeta}(s_1) \ll \log T. \tag{2.4}$$

From this estimate and corollary 2.6 we deduce that

$$\sum_{\rho} \Re \frac{1}{s_1 - \rho} \ll \log T, \tag{2.5}$$

where the sum is over all those zeros ρ for which $|\rho - (3/2 + it)| \leq 5/6$. Suppose that $1 - c/(2\log T) \leq \sigma \leq 1 + 1/\log T$. Then, again by corollary 2.6 we see that

$$\frac{\zeta'}{\zeta}(s) - \frac{\zeta'}{\zeta}(s_1) = \sum_{\rho} \left(\frac{1}{s - \rho} - \frac{1}{s_1 - \rho} \right) + O(\log T). \tag{2.6}$$

Since $|s - \rho| \asymp |s_1 - \rho|$ for all zeros ρ in the sum, it follows that

$$\frac{1}{s - \rho} - \frac{1}{s_1 - \rho} \ll \frac{1}{|s - \rho|^2} |s - s_1| \ll \Re \frac{1}{s_1 - \rho}.$$

Combining this with (2.4) and (2.5) in (2.6) we get $\zeta'(s)/\zeta(s) \ll \log \tau$.

For the case when $1 - c/(2\log \tau) < \sigma \leq 2$ and $|t| \leq 7/8$, we use (2.3). □

2.4 The Prime Number Theorem

Theorem 2.9. *There is a constant $c > 0$ such that*

$$\psi(x) = x + O\left(\frac{x}{e^{\sqrt{c \log x}}} \right), \tag{2.7}$$

$$\vartheta(x) = x + O\left(\frac{x}{e^{\sqrt{c \log x}}} \right), \tag{2.8}$$

and

$$\pi(x) = \mathrm{li}(x) + O\left(\frac{x}{e^{\sqrt{c \log x}}} \right), \tag{2.9}$$

uniformly for all $x \geq 2$.

Proof. Look at (2.1). From the truncated Perron formula we have

$$R \ll \sum_{x/2 < n < 2x} \Lambda(n) \min\left(1, \frac{1}{T|x - n|}\right) + \frac{x^{\sigma_0}}{T} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma_0}},$$

for $\sigma_0 > 1$ and $x > 0$. Note that $\Lambda(n) \leq \log n \ll \log x$.

$$\begin{aligned}\sum_{x/2<n<2x} \Lambda(n) \min\left(1, \frac{1}{T|x-n|}\right) &\ll (\log x)\left(1 + \frac{x}{T} \sum_{\substack{x/2<n<2x,\\ |x-n|\geq 1}} \frac{1}{|x-n|}\right) \\ &\ll (\log x)\left(1 + \frac{x}{T} \sum_{1\leq k\leq x} \frac{1}{k}\right) \\ &\ll \frac{x}{T}(\log x)^2.\end{aligned}$$

For the second term, note that the sum is $-\zeta'(\sigma_0)/\zeta(\sigma_0) \asymp 1/(\sigma_0 - 1)$ for $1 < \sigma_0 < 2$. Choosing $2 \leq T \leq x$ and $\sigma_0 = 1 + 1/\log x$, we obtain:

$$R \ll \frac{x}{T}(\log x)^2.$$

Put $\sigma_1 = 1 - c/\log T$ where c is small positive constant and let Γ denote the closed contour that consists of the line segments joining $\sigma_0 - iT, \sigma_0 + iT, \sigma_1 + iT$ and $\sigma_1 - iT$. Since $-\zeta'(s)/\zeta(s) = 1/(s-1) + O(1)$ for s close to 1 we know that $-\zeta'(s)/\zeta(s)$ has a simple pole at $s = 1$, but is otherwise analytic within Γ. Hence by the residue theorem:

$$-\frac{1}{2\pi i} \int_\Gamma \frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds = x.$$

Let T be large. From theorem 2.8 we have

$$-\frac{1}{2\pi i} \int_{\sigma_0+iT}^{\sigma_1+iT} \frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds \ll \frac{x^{\sigma_0}}{T} \log T(\sigma_0 - \sigma_1) \ll \frac{x}{T}.$$

Similarly we get

$$-\frac{1}{2\pi i} \int_{\sigma_1-iT}^{\sigma_0-iT} \frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds \ll \frac{x}{T}.$$

Again, by theorem 2.8, we see that

$$\begin{aligned}-\frac{1}{2\pi i} \int_{\sigma_1+iT}^{\sigma_1-iT} \frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds &\ll x^{\sigma_1} \log T \left(\int_{7/8}^{T} \frac{dt}{1+|t|} + \int_{-T}^{-7/8} \frac{dt}{1+|t|}\right) \\ &\quad + x^{\sigma_1} \int_{-7/8}^{7/8} \frac{dt}{|\sigma_1 - 1 + it|} \\ &\ll x^{\sigma_1} \log T \int_{-T}^{T} \frac{dt}{1+|t|} + x^{\sigma_1} \int_{-1}^{1} \frac{dt}{|\sigma_1 - 1|} \\ &\ll x^{\sigma_1} (\log T)^2 \ll x^{\sigma_1} (\log x)^2.\end{aligned}$$

On combining these estimates, we get:

$$-\frac{1}{2\pi i} \int_{\sigma_0-iT}^{\sigma_0+iT} \frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds \ll x + \frac{x}{T} + x^{\sigma_1} (\log x)^2.$$

Since $x/T \ll x/T(\log x)^2$, from (2.1) we see that

$$\begin{aligned}\psi(x) &= x + O\left(\frac{x}{T}(\log x)^2 + x^{\sigma_1}(\log x)^2\right) \\ &= x + O\left(x(\log x)^2\left(\frac{1}{T} + x^{-c/\log T}\right)\right).\end{aligned}$$

To minimize the error term an appropriate choice for T is when the two terms are equal. By taking logarithms, we get $T = e^{\sqrt{c\log x}}$. Therefore,

$$\psi(x) = x + O\left(\frac{x(\log x)^2}{e^{\sqrt{c\log x}}}\right).$$

Since $2\log\log x \ll (c\log x)^2$ (this is because $\log x \ll x^\theta$ for any $\theta > 0$), we conclude:

$$\psi(x) = x + O\left(\frac{x}{e^{\sqrt{c\log x}}}\right).$$

For (2.8), we use that fact that $\vartheta(x) = \psi(x) + O(\sqrt{x})$ for $x \geq 2$. This along with $\sqrt{x} \ll x\exp(-\sqrt{c\log x})$ gives (2.8).
For (2.9), using Riemann-Stieljes integral, we have

$$\pi(x) = \int_{2^-}^{x} \frac{1}{\log u} d\vartheta(u) = \mathrm{li}(x) + \int_{2^-}^{x} \frac{1}{\log u} du.$$

By integration by parts we see that the last integral is

$$\frac{\vartheta(u) - u}{\log u}\Big|_{2^-}^{x} + \int_{2}^{x} \frac{\vartheta(u) - u}{u(\log u)^2} du,$$

and by (2.8) it follows that this is $\ll x\exp(-\sqrt{c\log x})$ giving us (2.9). □

Chapter 3

Primes in Arithmetic Progression

3.1 Dirichlet Characters

Definition 3.1. *Let G be a finite abelian group. A Dirichlet character is a homomorphism $\chi : G \to \mathbb{S}^1$.*

Let $\hat{G}$ denote the set of all characters of G. We let χ_0 denote the trivial character, i.e., $\chi_0(g) = 1$ for all $g \in G$. Note that if $\chi \in \hat{G}$ then $\chi(e) = 1$. Let n denote the order of G. If $g \in G$ and $\chi \in \hat{G}$, then $\chi(g)^n = 1$ which implies that all the values taken by characters are nth roots of unity. In particular, this implies that $\hat{G}$ is finite, since there can be at most n^n such maps. If χ_1 and χ_2 are two characters of G, then we can define a product of characters $\chi_1\chi_2$ by $\chi_1\chi_2(g) = \chi_1(g)\chi_2(g)$. For $\chi \in \hat{G}$, let $\overline{\chi}$ be the character $\overline{\chi(g)}$. Then $\chi \cdot \overline{\chi} = \chi_0$, and we see that $\hat{G}$ is a finite abelian group with identity χ_0. The group of characters, $\hat{G}$ is called the **dual group** of G.

Lemma 3.2. *Suppose that G is cyclic of order n, say $G = (a)$. Then there are exactly n characters of G, namely $\chi_k(a^m) = e(km/n)$ for $1 \le k \le n$. Moreover,*

$$\sum_{g \in G} \chi(g) = \begin{cases} n & \text{if } \chi = \chi_0, \\ 0 & \text{otherwise,} \end{cases} \tag{3.1}$$

and

$$\sum_{\chi \in \hat{G}} \chi(g) = \begin{cases} n & \text{if } g = e, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2}$$

In this situation, $\hat{G}$ is cyclic and $\hat{G} = (\chi_1)$.

Proof. Suppose that $\chi \in \hat{G}$. We know that $\chi(a)$ is an nth root of unity, say $\chi(a) = e(k/n)$ for some $1 \le k \le n$. Hence $\chi(a^m) = \chi(a)^m = e(km/n)$. It is easy to see the remaining assertions. □

Lemma 3.3. *Suppose that G_1 and G_2 are finite abelian groups, and that $G = G_1 \otimes G_2$. If χ_i is a character of $G_i, i = 1, 2$ and $g \in G$ is written $g = (g_1, g_2)$, $g_i \in G_i$, then $\chi(g) = \chi_1(g)\chi_2(g)$ is a character of G. Conversely, if $\chi \in \hat{G}$, then there exist unique $\chi_i \in G_i$ such that $\chi(g) = \chi_1(g)\chi_2(g)$. The identities (3.1) and (3.2) holds for G if they hold for both G_1 and G_2.*

Proof. Left as an exercise to the reader. □

Theorem 3.4. *Let G be a finite abelian group. Then $\hat{G}$ is isomorphic to G, and (3.1) and (3.2) both hold.*

Proof. G is isomorphic to a direct product of cyclic groups, say

$$G \cong C_{n_1} \otimes C_{n_2} \otimes \cdots \otimes C_{n_r}.$$

The result follows from the lemmas above. □

For the pusposes of multiplicative number theory we take $G = (\mathbb{Z}/q\mathbb{Z})^*$, the multiplicative group of integers modulo q for some positive integer q. We have the following corollary.

Corollary 3.5. *The group $(\mathbb{Z}/q\mathbb{Z})^*$ has $\phi(q)$ Dirichlet characters. If χ is such a character, then*

$$\sum_{\substack{n=1 \\ (n,q)=1}}^{q} \chi(n) = \begin{cases} \phi(q) & \text{if } \chi = \chi_0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.3}$$

If $(n, q) = 1$, then

$$\sum_{\chi} \chi(n) = \begin{cases} \phi(q) & \text{if } n \equiv 1 (\text{mod } q), \\ 0 & \text{otherwise,} \end{cases} \tag{3.4}$$

where the sum is extended over the $\phi(q)$ Dirichlet characters χ modulo q.

Remark 3.6. *For our purposes it is convenient to define the Dirichlet characters (mod q) on all integers by setting $\chi(n) = 0$ for $(n, q) > 1$. Thus a Dirichlet character, χ is a totally multiplicative functions with period q which vanishes whenever $(n, q) > 1$.*

3.2 Dirichlet L-functions

Let χ be a character modulo q. For $\sigma > 1$ we put

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Since χ is totally multiplicative, by theorem 1.8 we have

$$L(s, \chi) = \prod_{p} \left(1 - \chi(p)p^{-s}\right)^{-1}$$

for $\sigma > 1$. Thus we see that

$$L(s, \chi_0) = \sum_{\substack{n=1 \\ (n,q)=1}}^{\infty} n^{-s} = \zeta(s) \prod_{p|q} \left(1 - p^{-s}\right) \tag{3.5}$$

for $\sigma > 1$. By (3.1) we see that if $\chi \neq \chi_0$, then

$$\left|\sum_{n\leq x} \chi(n)\right| \leq q$$

for any x, so that by theorem 1.5 the series $L(s,\chi)$ converges for $\sigma > 0$. Infact the function $L(s,\chi)$ is entire if $\chi \neq \chi_0$. But for $\sigma > 1$ we take logarithms and differentiate the Euler product for $L(s,\chi)$ to obtain the following.

Theorem 3.7. *If $\chi \neq \chi_0$, then $L(s,\chi)$ is analytic for $\sigma > 0$. On the other hand, the function $L(s,\chi_0)$ is analytic in this half-plane except for a simple pole at $s = 1$ with residue $\phi(q)/q$. In either case,*

$$\log L(s,\chi) = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{\log n}\chi(n)n^{-s} \tag{3.6}$$

for $\sigma > 1$, and

$$-\frac{L'}{L}(s,\chi) = \sum_{n=1}^{\infty} \Lambda(n)\chi(n)n^{-s}. \tag{3.7}$$

We now use the identity (3.4) to capture a prescribed residue class. If $(a,q) = 1$, then

$$\frac{1}{\phi(q)}\sum_{\chi} \overline{\chi(a)}\chi(n) = \begin{cases} 1 & \text{if } n \equiv a(\text{mod } q), \\ 0 & \text{otherwise} \end{cases} \tag{3.8}$$

where the sum is extended over all characters $\chi(\text{mod } q)$. Hence if $(a,q) = 1$, then

$$\sum_{\substack{n=1 \\ n\equiv a(q)}}^{\infty} \Lambda(n)n^{-s} = \frac{1}{\phi(q)}\sum_{n=1}^{\infty} \Lambda(n)n^{-s}\sum_{\chi} \overline{\chi(a)}\chi(n) = \frac{-1}{\phi(q)}\sum_{\chi} \overline{\chi(a)}\frac{L'}{L}(s,\chi) \tag{3.9}$$

for $\sigma > 1$. As $L(s,\chi_0)$ has a simple pole at $s = 1$, the function $\frac{L'}{L}(s,\chi_0)$ has a simple pole at $s = 1$. Thus the term arising from χ_0 on the right-hand side above is

$$\frac{1}{\phi(q)(s-1)} + O_q(1)$$

as $s \to 1^+$. This gives

$$\sum_{\substack{n=1 \\ n\equiv a(q)}}^{\infty} \Lambda(n)n^{-s} = \frac{1}{\phi(q)(s-1)} - \frac{1}{\phi(q)}\sum_{\chi\neq\chi_0} \overline{\chi(a)}\frac{L'}{L}(s,\chi) + O_q(1).$$

If the second sum is bounded when $s \to 1^+$ then it is easy to see that there are infinitely many primes $p \equiv a(\text{mod } q)$. This is because the sum $\sum_p p^{-ks}\log p$ converges for $k \geq 2$. Hence if there were finitely many such primes then the left-hand side would converge.

But $L(s,\chi)$ is analytic for $\sigma > 0$, so that $\frac{L'}{L}(s,\chi)$ is analytic xecept at zeros of $L(s,\chi)$. Hence

$$\lim_{s\to 1^+} \frac{L'}{L}(s,\chi) = \frac{L'}{L}(1,\chi)$$

for $\chi \neq \chi_0$, provided $L(1,\chi) \neq 0$. This is called the non-vanishing property of Dirichlet L-functions.

We have the following theorem.

Theorem 3.8 (Dirichlet)**.** *If χ is a character modulo q with $\chi \neq \chi_0$, then $L(1,\chi) \neq 0$.*

As a corollary we have.

Corollary 3.9 (Dirichlet's theorem)**.** *If $(a,q) = 1$, then there are infinitely many primes $p \equiv a (mod\ q)$.*

We call a character real if all its values are real and complex otherwise. A character is quadratic if it has order 2 in the cahracter group: $\chi^2 = \chi_0$ but $\chi \neq \chi_0$. Thus a quadratic character is real, and a real character is either quadratic or principal.

To prove theorem 3.8 we need a result by Landau

Theorem 3.10 (Landau)**.** *Let $\alpha(s) = \sum a_n n^{-s}$ be a Dirichlet series with non-negative real coefficients. Suppose σ_a is the abscissa of of absolute convergence for $\alpha(s)$. Then α cannot be extended to a holomorphic function on any neighbourhood of $s = \sigma_a$.*

Proof. Suppose on the contrary that α extends to a holomorphic function on the disc $|s - \sigma_a| < \varepsilon$. Pick a real number $c \in (\sigma_a, \sigma_a + \varepsilon/2)$, and write

$$\begin{aligned}
\alpha(s) &= \sum_{n=1}^{\infty} a_n n^{-c} n^{c-s} \\
&= \sum_{n=1}^{\infty} a_n n^{-c} \exp\left((c-s)\log n\right) \\
&= \sum_{n=1}^{\infty} \sum_{i=0}^{\infty} \frac{a_n n^{-c} (\log n)^i}{i!} (c-s)^i.
\end{aligned}$$

Since all coefficients in this double series are nonnegative, everything must converge absolutely in the disc $|s - c| < \varepsilon/2$. In particular, when viewed as a power series in $(c-s)$, this must give the Taylor series for α around $s = c$. Since α is holomorphic in the disc $|s - c| < \varepsilon/2$, the Taylor series converges there; in particular, it converges for some real number $\sigma_a' < \sigma_a$. But now we can run the argument backwards to deduce that the original Dirichlet series converges absolutely for $s = \sigma_a'$, which implies that the abscissa of absolute convergence is at most σ_a'. This contradicts the definition of σ_a'. □

Proof of non-vanishing 3.8. We treat the quadratic and complex characters separately.
Case 1: Complex χ. From (3.6) we have

$$\prod_{\chi} L(s,\chi) = \exp\left(\sum_{\chi} \sum_{n=2}^{\infty} \frac{\Lambda(n)}{\log n} \chi(n) n^{-s}\right)$$

for $\sigma > 1$. By (3.4) this is

$$= \exp\left(\phi(q) \sum_{\substack{n=2 \\ n\equiv 1(q)}}^{\infty} \frac{\Lambda(n)}{\log n} n^{-s}\right).$$

If we take $s = \sigma > 1$, then the sum above is non-negative real number, and hence we see that

$$\prod_{\chi} L(\sigma, \chi) \geq 1$$

for $\sigma > 1$. Now $L(s, \chi_0)$ has a simple pole at $s = 1$, but the other $L(s, \chi)$ are analytic at $s = 1$. Thus $L(1, \chi) = 0$ can hold for at most one χ, since otherwise the poles would cancel making the above product tend to 0 as $\sigma \to 1^+$. If χ is a character modulo q, then $\overline{\chi}$ is a character modulo q, and $\chi \neq \overline{\chi}$ in this case. Moreover $\overline{L(s, \chi)} = L(\overline{s}, \overline{\chi})$ by Schwarz reflection principle, so that $L(1, \overline{\chi}) = 0$ is $L(1, \chi) = 0$. Consequently $L(1, \chi) \neq 0$ for complex characters.

Case 2: Quadratic χ. Let $r(n) = \sum_{d|n} \chi(d)$. Thus $\alpha(s) = \sum_{n=1}^{\infty} r(n) n^{-s} = \zeta(s) L(s, \chi)$ for $\sigma > 1$. Also note that $r(n)$ is multiplicative and

$$r(p^\alpha) = \begin{cases} 1 & \text{if } p \mid q, \\ \alpha + 1 & \text{if } \chi(p) = 1, \\ 1 & \text{if } \chi(p) = -1 \text{ and } 2 \mid \alpha, \\ 0 & \text{if } \chi(p) = -1 \text{ and } 2 \nmid \alpha. \end{cases}$$

Here $r(n) \geq 0$ for all n, and $r(n^2) \geq 1$ for all n. Suppose that $L(1, \chi) = 0$. Then $\zeta(s) L(s, \chi)$ is analytic for $\sigma > 0$, and by Landau's theorem, the abscissa of absolute convergence for α cannot be positive. Therefore the series $\alpha(s)$ converges for $\sigma > 0$. But this is false since

$$\sum_{n=1}^{\infty} r(n) n^{-1/2} \geq \sum_{n=1}^{\infty} r(n^2) n^{-1} \geq \sum_{n=1}^{\infty} n^{-1} = \infty.$$

Hence $L(1, \chi) \neq 0$. Since $L(\sigma, \chi) > 0$ for $\sigma > 1$ when χ is quadratic, we see that $L(1, \chi) > 0$ in this case. □

3.3 PNT in Arithmetic Progression

Lemma 3.11. *Let χ be a non-principle character modulo q, and suppose that $\delta > 0$ is fixed. Then*

$$L(s, \chi) \ll (1 + (q\tau)^{1-\sigma}) \min\left(\frac{1}{|\sigma - 1|}, \log q\tau\right)$$

uniformly for $\delta \leq \sigma \leq 2$. In particular taking $\delta = 1/2$ we have a crude bound

$$L(s, \chi) \ll q\tau.$$

Proof. We prove the second bound which will be relevant to the rest of the chapter. A full proof of this theorem can be found in [2] chapter 10.

Let $S = \sum_{n \leq x} \frac{\chi(n)}{n^s}$. Using the Riemann-Stieljes integral we have

$$\begin{aligned} L(s,\chi) &= \sum_{n \leq x} \frac{\chi(n)}{n^s} + \int_{x^-}^{\infty} \frac{d(S(u))}{u^s} \\ &= \sum_{n \leq x} \frac{\chi(n)}{n^s} - \frac{S(x)}{x^s} + s\int_x^{\infty} \frac{S(u)}{u^{s+1}} du \ll x + \frac{q}{x^\sigma} + q\tau. \end{aligned}$$

Taking $x = q\tau$, we get the desired bound. □

3.3.1 Approximate Partial Fraction Decomposition of L'/L

Let $E_0(\chi)$ be 0 for non trivial characters and 1 otherwise.

Lemma 3.12. *If χ is a character (mod q) and $5/6 \leq \sigma \leq 2$, then*

$$-\frac{L'}{L}(s,\chi) = \frac{E_0(\chi)}{s-1} - \sum_{\rho} \frac{1}{s-\rho} + O(\log q\tau)$$

where the sum if over all zeros ρ of $L(s,\chi)$ for which $|\rho - (3/2 + it)| \leq 5/6$.

Proof. When χ is non-principle we apply (2.5) to the function

$$f(z) = L\left(z + \left(\frac{3}{2} + it\right), \chi\right)$$

with $R = 5/6$ and $r = 2/3$. By lemma 3.11 we can take $M = Cq\tau$ for a suitable constant C, and by the Euler product we see that

$$|f(0)| = \left|L\left(\frac{3}{2} + it, \chi\right)\right| = \prod_p \left|1 - \chi(p)p^{-\frac{3}{2}-it}\right|^{-1} \geq \prod_p \left|1 + p^{-\frac{3}{2}}\right|^{-1} \gg 1.$$

$$\begin{aligned} \frac{f'}{f}(z) = \frac{L'}{L}\left(z + \frac{3}{2} + it, \chi\right) &= \sum_{\substack{f(\eta)=0 \\ |\eta| \leq R}} \frac{1}{z-\eta} + O(\log q\tau) \\ \implies -\frac{L'}{L}(s,\chi) &= -\sum_{\substack{|\rho-(3/2+it)| \leq 5/6 \\ L(\rho,\chi)=0}} \frac{1}{s-\rho} + O(\log q\tau). \end{aligned}$$

When $\chi = \chi_0$, we know that

$$L(s,\chi_0) = \zeta(s) \prod_{p|q} \frac{\log p}{p^s - 1}.$$

This shows that the roots of $L(s,\chi_0)$ in $|\rho-(3/2+it)|\leq 5/6$ are precisely the zeros of $\zeta(s)$ in that region because the zeros of the finite product are on the imaginary axis. Now on taking logarithms and differentiating we see

$$\begin{aligned}\frac{L'}{L}(s,\chi_0) &= \frac{\zeta'}{\zeta}(s)+\sum_{p|q}\frac{\log p}{p^s-1}\\ &= \frac{\zeta'}{\zeta}(s)+O(\log q) \quad \text{for } \sigma\geq 5/6.\end{aligned}$$

By corollary 2.6 the result follows for $|t|>7/8$.
For $|t|\leq 7/8$, we apply lemma 2.5 to the function $f(z)=(z+1/2+it)\,\zeta\,(z+3/2+it)$ which completes the proof. □

Lemma 3.13. *If $\sigma>1$, then*

$$\Re\left(-3\frac{L'}{L}(\sigma,\chi_0)-4\frac{L'}{L}(\sigma+it,\chi)-\frac{L'}{L}(\sigma+2it,\chi^2)\right)\geq 0.$$

Proof. By the Dirichlet series expansion for L'/L we see that the left-hand side of the above is equal to

$$\Re\sum_{\substack{n=1\\(n,q)=1}}^{\infty}\frac{\Lambda(n)}{n^\sigma}\left(3+4\chi(n)n^{-it}+\chi^2(n)n^{-2it}\right).$$

The quantity $\chi(n)n^{-it}$ is unimodular when $(n,q)=1$, so for such n there is a real number θ_n such that $\chi(n)n^{-it}=e^{i\theta_n}$. Thus the above is

$$\sum_{\substack{n=1\\(n,q)=1}}^{\infty}\frac{\Lambda(n)}{n^\sigma}(3+4\cos\theta_n+\cos 2\theta_n).$$

This is non-negative because $3+4\cos\theta+\cos 2\theta=2(1+\cos\theta)^2\geq 0$ for all θ. □

3.4 A Zero-Free Region for Dirichlet's L Functions.

Theorem 3.14. *There is an absolute constant $c>0$ such that if χ is a dirichlet character modulo q, then the region*

$$R_q=\{s:\sigma>1-1/\log q\tau\}$$

contains no zero of $L(s,\chi)$ unless χ is quadratic, in which case $L(s,\chi)$ has at most one, necessarily real, zero $\beta<1$ in R_q.

Proof. The case when $\chi=\chi_0$ is clear by the analytic continuation and zero-free region of ζ. So we may assume that χ is non-principle. Also, the Euler product for $L(s,\chi)$ is absolutely convergent for $\sigma>1$, and hence $L(s,\chi)\neq 0$ for such s. Thus it suffices to consider a zero $\rho_0=\beta_0+i\gamma_0$ of $L(s,\chi)$ with $12/13\leq\beta_0\leq 1$. The lower bound is chosen so that the above lemmas are applicable. We consider several cases.

Case 1. (Complex χ) If $\sigma > 1$ and ρ is a zero of $L(s,\chi)$, then $\Re(s-\rho) > 0$ and hence $\Re(1/(s-\rho)) > 0$. Thus by the partial fraction lemma 3.12, if $0 < \delta \leq 1$, then

$$\begin{aligned} -\Re\frac{L'}{L}(1+\delta,\chi_0) &\leq \frac{1}{\delta} + c_1 \log q, \\ -\Re\frac{L'}{L}(1+\delta+i\gamma_0,\chi) &\leq \frac{-1}{1+\delta-\beta_0} + c_1 \log q\tau, \\ -\Re\frac{L'}{L}(1+\delta+2i\gamma_0,\chi^2) &\leq c_1 \log q\tau \end{aligned} \tag{3.10}$$

for some absolute constant c_1. Taking the appropriate linear combination and using lemma 3.13 we see that the resulting left-hand side is non-negative. That is,

$$\frac{3}{\delta} - \frac{4}{1+\delta-\beta_0} + c_2 \log q\tau \geq 0$$

for some absolute constant c_2. Note that $\beta_0 < 1$ as otherwise taking $\delta \to 0^+$ in the above inequality, we get a contradiction. Therefore, taking $\delta = 6(1-\beta_0)$, (mainly to simplify the calculations) we get

$$1 - \beta_0 \geq \frac{1}{14 c_2 \log q\tau}.$$

Hence $\rho_0 \notin R_q$ is c is sufficiently small.

Case 2. (Quadratic χ, $|\gamma_0| \geq 6(1-\beta_0)$) By non-vanishing of $L(1,\chi)$ $\gamma_0 \neq 0$. Hence we can proceed as in case 1 except for the third inequality where $\chi^2 = \chi_0$ which introduces an extra term $1/(s-1)$ from lemma 3.12. So we replace the third inequality with a weaker inequality

$$-\Re\frac{L'}{L}(1+\delta+2i\gamma_0,\chi^2) \leq \frac{\delta}{\delta^2+4\gamma_0^2} + c_1 \log q(2|\gamma_0|+4).$$

Again taking the appropriate linear combination we get

$$\frac{3}{\delta} - \frac{4}{1+\delta-\beta_0} + \frac{\delta}{\delta^2+4\gamma_0^2} + c_2 \log q\tau \geq 0.$$

So if $\beta_0 = 1$, then taking $\delta \to 0^+$ we get a contradiction, therefore $\beta_0 < 1$. Also $|\gamma_0| \geq 6(1-\beta_0)$ implies

$$-\Re\frac{L'}{L}(1+\delta+2i\gamma_0,\chi^2) \leq \frac{\delta}{\delta^2+144(1-\beta_0)^2} + c_1 \log q(2|\gamma_0|+4).$$

Taking $\delta = 6(1-\beta_0)$ we see

$$\frac{1}{1-\beta_0}\left(\frac{3}{6} - \frac{4}{7} + \frac{6}{180}\right) + c_2 \log q\tau \geq 0.$$

The factor in large parantheses above is $-4/105 < -1/27$, so

$$1 - \beta_0 \geq \frac{1}{27 c_2 \log q\tau}.$$

Case 3. (Quadratic χ, $0 < |\gamma_0| \leq 6(1-\beta_0)$) Since $L(s,\chi)$ is real when s us real, it follows by Schwarz reflection principle that $L(\overline{\rho},\chi) = 0$. Hence by lemma 3.12 we see that if $1 < \sigma \leq 2$, then

$$-\Re\frac{L'}{L}(\sigma,\chi) \leq -\Re\frac{1}{\sigma-\rho_0} - \Re\frac{1}{\sigma-\overline{\rho_0}} + c_1\log 4q = \frac{-2(\sigma-\beta_0)}{(\sigma-\beta_0)^2} + c_1\log 4q$$

$$\leq \frac{-2(\sigma-\beta_0)}{(\sigma-\beta_0)^2 + 36(1-\beta_0)^2} + c_1\log 4q. \tag{3.11}$$

Now observe that if $\sigma > 1$, then

$$-\frac{L'}{L}(\sigma,\chi_0) - \frac{L'}{L}(\sigma,\chi) = \sum_{\substack{n=1\\(n,q)=1}}^{\infty} \frac{\Lambda(n)}{n^\sigma}(1+\chi(n)) \geq 0. \tag{3.12}$$

We put $\delta = a(1-\beta_0)$ and combine (3.11) with the first inequality in (3.10) in the above to get

$$\frac{1}{1-\beta_0}\left(\frac{1}{a} - \frac{2(a+1)}{(a+1)^2+36}\right) + c_2\log 4q \geq 0.$$

The factor in the large parantheses is $\approx -1/a$ as $a \to \infty$. So taking $a = 13$, this factor becomes $-33/754$ which is less than $-1/27$, and hence

$$1-\beta_0 \geq \frac{1}{27c_2\log 4q}.$$

Case 4. (Quadratic χ, real zeros, i.e., $\gamma_0 = 0$) If β_0 is a real zero of $L(s,\chi)$, then $\beta < 1$. Suppose that $\beta_0 \leq \beta_1 < 1$ are two such zeros. Then by lemma 3.12, ($\sigma > 1$)

$$-\Re\frac{L'}{L}(\sigma,\chi) \leq -\frac{1}{\sigma-\beta_0} - \frac{1}{\sigma-\beta_1} + c_1\log 4q$$
$$\leq -\frac{2}{\sigma-\beta_0} + c_1\log 4q.$$

Combining the first part of (3.10) and the above in (3.12) with $\sigma = 1+\delta = 1+a(1-\beta_0)$, we find that

$$\frac{1}{1-\beta_0}\left(\frac{1}{a} - \frac{2}{a+1}\right) + c_2\log 4q \geq 0.$$

On taking $a = 2$ we deduce that

$$1-\beta_0 \geq \frac{1}{6c_2\log 4q}.$$

This completes the proof. □

3.5 Bounds on L'/L in the Zero-Free Region

Theorem 3.15. *Let χ be a non-principle character modulo q, let c be the constant in previous theorem, and suppose that $\sigma \geq 1 - c/(2\log q\tau)$. If $L(s,\chi)$ has no exceptional zero, or if β_1 is an exceptional zero of $L(s,\chi)$ with $|s-\beta_1| \geq 1/\log q$, then*

$$\frac{L'}{L}(s,\chi) \ll \log q\tau. \tag{3.13}$$

Alternatively, if β_1 is an exceptional zero of $L(s,\chi)$ and $|s-\beta_1| \leq 1/\log q$, then

$$\frac{L'}{L}(s,\chi) = \frac{1}{s-\beta_1} + O(\log q) \quad (s \neq \beta_1). \tag{3.14}$$

Proof. If $\sigma > 1$, then

$$\left|\frac{L'}{L}(s,\chi)\right| \leq \sum_{n\geq 1} \Lambda(n)n^{-\sigma} = -\frac{\zeta'}{\zeta}(\sigma) \ll \frac{1}{\sigma-1}.$$

Hence (3.13) is obvious if $\sigma \geq 1 + 1/\log q\tau$. Let $s_1 = 1 + 1/\log q\tau + it$. Then

$$\frac{L'}{L}(s_1,\chi) \ll \log q\tau.$$

From this and lemma 3.12 it follows that

$$\sum_{\rho} \frac{1}{s_1-\rho} \ll \log q\tau \tag{3.15}$$

where the sum is over those zeros of $L(s,\chi)$ for which $|\rho - (3/2+it)| \leq 5/6$. Hence

$$\sum_{\rho} \frac{1}{s-\rho} = \sum_{\rho}\left(\frac{1}{s-\rho} - \frac{1}{s_1-\rho}\right) + O(\log q\tau).$$

Suppose that $1 - c/(2\log q\tau) \leq \sigma \leq 1 + 1/\log q\tau$ and that $|s-\beta_1| \geq 1/\log q$ if $L(s,\chi)$ has an exceptional zero β_1. Since $|s-\rho| \asymp |s_1-\rho|$ for all zeros ρ, it follows that

$$\frac{1}{s-\rho} - \frac{1}{s_1-\rho} = \frac{1 + 1/\log q\tau - \sigma}{(s-\rho)(s_1-\rho)} \ll \frac{1}{|s_1-\rho|^2 \log q\tau} \ll \Re\frac{1}{s_1-\rho}.$$

Since there are only constant number of roots, ρ, on summing this over all the zeros ρ and appealing to (3.15), (3.13) follows.

Now suppose that $L(s,\chi)$ has an exceptional zero β_1 such that $|s-\beta_1| \leq 1/\log q$. Then $1 - c/(2\log 4q) \leq \sigma \leq 1 + 1/\log q$, so by lemma 3.12,

$$\frac{L'}{L}(s,\chi) = \frac{1}{s-\beta_1} + \sum_{\substack{|\rho-(3/2+it)| \\ \rho\neq\beta_1}} \frac{1}{s-\rho} + O(\log q).$$

As in the previous case the sum is $\ll \log q\tau \ll \log q$ because τ is bounded in this case, we get the desired result. □

3.6 Proof of PNT in Arithmetic Progressions

We set up some notations for the prime number theorem. For $(a,q)=1$ we set

$$\pi(x;q,a)=\sum_{\substack{p\le x\\ p\equiv a(q)}}1,\quad \vartheta(x;q,a)=\sum_{\substack{p\le x\\ p\equiv a(q)}}\log p,\quad \psi(x;q,a)=\sum_{\substack{n\le x\\ n\equiv a(q)}}\Lambda(n),$$

and correspondingly for any Dirichlet character χ we put

$$\pi(x,\chi)=\sum_{p\le x}\chi(p),\quad \vartheta(x,\chi)=\sum_{p\le x}\chi(p)\log p,\quad \psi(x,\chi)=\sum_{n\le x}\chi(n)\Lambda(n).$$

By multiplying both sides of (3.8) by $\Lambda(n)$ and summing over $n\le x$, we see that

$$\psi(x;q,a)=\frac{1}{\phi(q)}\sum_{\chi}\overline{\chi(a)}\psi(x,\chi), \tag{3.16}$$

and similarly for $\pi(x;q,a)$ and $\vartheta(x;q,a)$.

Theorem 3.16. *There is a constant $c_1>0$ such that if $q\le \exp(2c_1\sqrt{\log x})$, then*

$$\psi(x,\chi)=E_0(\chi)+O(x\exp(-c_1\sqrt{\log x})) \tag{3.17}$$

when $L(s,\chi)$ has no exceptional zero, but

$$\psi(x,\chi)=-\frac{x^{\beta_1}}{\beta_1}+O(x\exp(-c_1\sqrt{\log x})) \tag{3.18}$$

when $L(s,\chi)$ has an exceptional zero β_1.

Proof. By Perron's Formula we have

$$\psi(x,\chi)=-\frac{1}{2\pi i}\int_{\sigma_0-iT}^{\sigma_0+iT}\frac{L'}{L}(s,\chi)\frac{x^s}{s}ds+R$$

where $\sigma_0>1$ and

$$R\ll\sum_{x/2<n<2x}\Lambda(n)\min\left(1,\frac{x}{T|x-n|}\right)+\frac{x^{\sigma_0}}{T}\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^{\sigma_0}}.$$

As in the proof of the prime number theorem we suppose that $2\le T\le x$ and set $\sigma_0=1+1/\log x$. Thus

$$R\ll\frac{x}{T}(\log x)^2$$

as before. Let $\mathcal{C}$ denote the closed contour consisting of line segments joining the points $\sigma_0-iT,\sigma_0+iT,\sigma_1+iT$ and σ_1-iT, but now the choice of σ_1 is a little more complicated, since we want to ensure that $\mathcal{C}$ does not pass too closely to an exceptional zero.

Case 1. (There is no exceptional zero.) In this case we take $\sigma_1 = 1 - c/(5\log qT)$ where c is the constant in theorem 3.14. If χ is a non-principle character, then the integrand is analytic on and inside $\mathcal{C}$, but if $\chi = \chi_0$, then it has a pole at $s = 1$ with residue x. Hence

$$-\frac{1}{2\pi i}\int_{\mathcal{C}} \frac{L'}{L}(s,\chi)\frac{x^s}{s}ds = E_0(\chi)x. \tag{3.19}$$

Following the proof of the prime number theorem we find that

$$\psi(x,\chi) - E_0(\chi) \ll x(\log x)^2\left(\frac{1}{T} + \exp\left(\frac{-c\log x}{5\log qT}\right)\right). \tag{3.20}$$

Case 2. (There is an exceptional zero β_1, ans satisfies $\beta_1 \geq 1 - c/(4\log qT)$). In this case we take $\sigma_1 = 1 - c/(3\log qT)$. The integrand in (3.19) now has a pole inside $\mathcal{C}$ at β_1, so the left-hand side of (3.19) has the value $-x^{\beta_1}/\beta_1$. Otherwise, the estimates proceed as before, and we find that

$$\psi(x,\chi) = -\frac{x^{\beta_1}}{\beta_1} + O\left(x(\log x)^2\left(\frac{1}{T} + \exp\left(\frac{-c\log x}{5\log qT}\right)\right)\right). \tag{3.21}$$

Case 3. There is an exceptional zero β_1 satisfying $\beta_1 < 1 - c/(4\log qT)$. We proceed as in case 1, and so we obtain (3.20). Note that in the current case

$$\frac{x^{\beta_1}}{\beta_1} \ll x\exp\left(\frac{-c\log x}{5\log qT}\right)$$

which gives (3.21).

Thus we have established (3.20) if there is no exceptional zero, and (3.21) is there is one. Now we only need to observe that if $c_1 = \sqrt{c/20}$, $q \leq \exp(2c_1\sqrt{\log x})$ and if $T = \exp(2c_1\sqrt{\log x})$, then (3.20) implies (3.17) and (3.21) gives (3.18). □

As a corollary we obtain

Corollary 3.17 (Page)**.** *Let c_1 be the same constant as in theorem 3.16. If $(a,q) = 1$, then*

$$\psi(x;q,a) = \frac{x}{\phi(q)} + O(x\exp(-c_1\sqrt{\log x})) \tag{3.22}$$

when there is no exceptional character modulo q, and

$$\psi(x;q,a) = \frac{x}{\phi(q)} - \frac{\chi_1(a)x^{\beta_1}}{\phi(q)\beta_1} + O(x\exp(-c_1\sqrt{\log x})) \tag{3.23}$$

when there is an exceptional character χ_1 modulo q and β_1 is the concomitant zero.

Proof. For $q \leq \exp(2c_1\sqrt{\log x})$, substitute the estimates (3.17) and (3.18) in (3.16). But when $q > \exp(2c_1\sqrt{\log x})$, the stated estimates are worse than trivial bound. To see that, note that the largest term in $\psi(x;q;a)$ is $\leq \log x$, and the number of terms is $\leq x/q$, so it is immediate that

$$\psi(x;q,a) \leq (x/q+1)\log x \ll \exp(-c_1\sqrt{\log x})$$

when $q > \exp(2c_1\sqrt{\log x})$. □

Corollary 3.18. *(Prime Number Theorem for Arithmetic Progressions) Let c_1 be the constant in theorem 3.16. If $(a,q)=1$, then*

$$\vartheta(x;q,a)=\frac{x}{\phi(q)}+O(x\exp(-c_1\sqrt{\log x})) \tag{3.24}$$

and

$$\pi(x;q,a)=\frac{li(x)}{\phi(q)}+O(x\exp(-c_1\sqrt{\log x})) \tag{3.25}$$

when there is no exceptional character modulo q, but

$$\vartheta(x;q,a)=\frac{x}{\phi(q)}-\frac{\chi_1(a)x^{\beta_1}}{\phi(q)\beta_1}+O(x\exp(-c_1\sqrt{\log x})) \tag{3.26}$$

and

$$\pi(x;q,a)=\frac{li(x)}{\phi(q)}-\frac{\chi_1(a)li(x^{\beta_1})}{\phi(q)}+O(x\exp(-c_1\sqrt{\log x})) \tag{3.27}$$

when there is an exceptional character χ_1 modulo q and β_1 is the concominant zero.

Proof. Since

$$0\le\psi(x;q,a)-\vartheta(x;q,a)\le\psi(x)-\vartheta(x)\ll x^{1/2},$$
$$\implies \vartheta(x;q,a)=\psi(x;q,a)+O(x^{1/2}).$$

Now the assertion concerning $\vartheta(x;q,a)$ follow immediately from corollary 3.17.
For $\pi(x;q,a)$, similar to the proof of prime number theorem, via Riemann-Stiljes integration we see that

$$\pi(x;q,a)=\int_{2^-}^{x}\frac{1}{\log u}d\vartheta(u;q,a)=\frac{\mathrm{li}(x)}{\phi(q)}+\int_{2^-}^{x}\frac{1}{\log u}d(\vartheta(u;q,a)-u/\phi(q)).$$

By integration by parts we find that this last integral is

$$\frac{\vartheta(u;q,a)-u/\phi(q)}{\log u}\Big|_{2^-}^{x}-\int_{2}^{x}\frac{\vartheta(u;q,a)-u/\phi(q)}{(u\log u)^2}du.$$

If there is no exceptional zero, then by using (3.24), the numerator in the integrand is $\ll u\exp(-c_1\sqrt{\log u})\ll x\exp(-c_1\sqrt{\log x})$, so we obtain (3.25). If there is an exceptional character χ_1, then (3.26) gives

$$\begin{aligned}\pi(x;q,a)&=\frac{\mathrm{li}(x)}{\phi(q)}-\frac{\chi_1(a)}{\phi(q)}\int_{2^-}^{x}\frac{1}{\log u}\,d\left(\frac{u^{\beta_1}}{\beta_1}\right)+\int_{2^-}^{x}\frac{1}{\log u}\,d\big(O(u\exp(-c_1\sqrt{\log u}))\big)\\&=\frac{\mathrm{li}(x)}{\phi(q)}-\frac{\chi_1(a)\mathrm{li}(x^{\beta_1})}{\phi(q)}+O(x\exp(-c_1\sqrt{\log x}))\\&\qquad-\int_{2}^{x}O(u\exp(-c_1\sqrt{\log u}))\,d\left(\frac{1}{\log u}\right)\\&=\frac{\mathrm{li}(x)}{\phi(q)}-\frac{\chi_1(a)\mathrm{li}(x^{\beta_1})}{\phi(q)}+O(x\exp(-c_1\sqrt{\log x})).\end{aligned}$$

This completes the proof. □

3.7 Siegel-Walfisz theorem

We state the following theorem due to Siegel without proof. A proof of this theorem can be found in [2] chapter 11.

Theorem 3.19 (Siegel)**.** *For each positive number ε there is a positive constant $C(\varepsilon)$ such that if χ is a quadratic character modulo q, then*

$$L(1,\chi) > C(\varepsilon)q^{-\varepsilon}.$$

Theorem 3.20. *Let c_1 be the same constant as in theorem 3.16. For any positive A there is an $x_0(A)$ such that if $q \leq (\log x)^A$, then*

$$\phi(x,\chi) = E_0(\chi)x + O(e\exp(-c_1\sqrt{\log x}))$$

for $x \geq x_0(A)$.

Proof. Suppose that χ is quadratic and that $L(s,\chi)$ has an exceptional zero β_1. Then

$$x\exp(-(1-\beta_1)\log x) \leq x\exp(-C(\varepsilon)q^{-\varepsilon}\log x)$$

by Siegel's theorem. Since $q \leq (\log x)^A$, the above is

$$\leq x\exp(-C(\varepsilon)(\log x)^{1-A\varepsilon}).$$

Taking $\varepsilon = 1/(3A)$ the above becomes

$$\leq x\exp(-c_1\sqrt{\log x})$$

provided that $x \geq x_0 = \exp((c_1/C(\varepsilon))^6)$. □

We get the following result which we state as a corollary.

Corollary 3.21. *Let c_1 be the constant in theorem 3.16, and suppose that A is given, $A > 0$. If $q \leq (\log x)^A$ and $(a,q) = 1$, then*

$$\vartheta(x;q,a) = \frac{x}{\phi(q)} + O_A(x\exp(-c_1\sqrt{\log x}))$$

and

$$\pi(x;q,a) = \frac{li(x)}{\phi(q)} + O_A(x\exp(-c_1\sqrt{\log x})).$$

Chapter 4

The Ternary Goldbach Conjecture

4.1 Introduction

In 1742 in an exchange of letters with Euler, Goldbach conjectured that every even number greater than 2 is a sum of two primes (remains a conjecture) and a variant of this problem, the ternary Goldbach conjecture or the three-primes problem, which asserts that every odd number greater than 5 can be expressed as a sum of three prime numbers.

Although the first conjecture is still unsolved, in 1922 Hardy and Littlewood used the circle method, and showed assuming the Generalized Riemann hypothesis, that every sufficiently large odd integer is a sum of three primes.

This was first proven unconditionally by I. M. Vinogradov in 1937. The conjecture was finally resolved in 2013 by H. A. Helfgott. He proved that the conjecture is true for all odd integers greater than 5.

We follow the Vinogradov's proof. Instead of considering the number of representations of n as sum of three primes, Vinogradov considered a related quantity

$$r(n) = \sum_{k_1+k_2+k_3=n} \Lambda(k_1)\Lambda(k_2)\Lambda(k_3),$$

which is a weighted sum of the number of representations of n as a sum of three prime powers.

Let

$$r'(n) = \sum_{\substack{p_1+p_2+p_3=n \\ p_1,p_2,p_3 \text{ are prime}}} 1.$$

Note that $r(n) > 0$ does not imply $r'(n) > 0$. But

$$\begin{aligned} r(n) - r'(n) &\ll \sum_{\substack{p^l+k_2+k_3=n \\ l\geq 2}} \Lambda(p^l)\Lambda(k_2)\Lambda(k_3) \\ &\ll (\log n)^2 \sum_{\substack{p^l+k_2+k_3=n \\ l\geq 2}} \Lambda(p^l) \\ &\ll (\log n)^2 \sum_{\substack{p^l\leq n \\ l\geq 2}} \Lambda(p^l) \sum_{k_2+k_3=n-p^l} 1 \end{aligned}$$

$$\ll n(\log n)^2 \sum_{\substack{p^l \leq n \\ l \geq 2}} \log p$$
$$\ll n(\log n)^2 \sum_{2 \leq l \leq \log_2 n} \vartheta(n^{1/l}),$$

where $\vartheta(x) = \sum_{p \leq x} \log p$. Now, via partial summation we get

$$\vartheta(x) = \int_{2^-}^{x} \log u \, d(\pi(u)) = \pi(x) \log x - \int_2^x \frac{\pi(u)}{u} du.$$

Therefore, the prime number theorem gives $\vartheta(x) \ll x$. This gives

$$n(\log n)^2 \sum_{2 \leq l \leq \log_2 n} \vartheta(n^{1/l}) \ll n^{3/2} (\log n)^3.$$

Thus $r(n) - r'(n) \ll n^{3/2}(\log n)^3$.

theorem 4.1, stated below, proves that $r(n) \gg n^2$. This together with our calculations above shows that if for large n, $r(n) > 0$ then $r'(n) > 0$, proving the ternary Goldbach conjecture for all sufficiently large n.

Theorem 4.1. *(Vinogradov 1937) For any fixed $A > 0$,*

$$r(N) = \frac{1}{2} \mathfrak{G}(N) N^2 + O\left(\frac{N^2}{(\log N)^A} \right),$$

where

$$\mathfrak{G}(N) = \left(\prod_{p \mid N} \left(1 - \frac{1}{(p-1)^2} \right) \prod_{p \nmid N} \left(1 + \frac{1}{(p-1)^3} \right) \right).$$

The above is not very useful when N is even as $\mathfrak{G}(N) = 0$. But when N is odd, then

$$\prod_p \left(1 - \frac{1}{(p-1)^2} \right) < \mathfrak{G}(N) < \prod_p \left(1 + \frac{1}{(p-1)^3} \right).$$

The infinite products on both sides are convergent. So indeed theorem 4.1 gives $r(N) \gg N^2$.

In additive problems it is appropriate to use a generating function of an exponential sum. So let $e^{2\pi i \alpha} = e(\alpha)$, and consider the exponential sum

$$S(\alpha) = \sum_{k \leq N} \Lambda(k) e(k\alpha),$$

we see that

$$S(\alpha)^3 \sum_n r''(n) e(n\alpha),$$

where r'' is defined in the same was as $r(n)$ with the further restriction that the k_i do not exceed N. Hence $r''(n) = r(n)$ for $n \leq N$. By the Fourier coefficient formula we have

$$r(N) = \int_0^1 S(\alpha)^3 e(-N\alpha) d\alpha. \tag{4.1}$$

We observe that if $\alpha = a/q$ is a rational number with $(a, q) = 1$. For each power of a prime k we write $ka = lq + r$, where $(r, q) = 1$ for most of the values taken by r. Grouping the terms in $S(\alpha)$ with the same residue class, we find

$$S(\alpha) \approx \sum_{\substack{1 \leq r \leq q \\ (r,q)=1}}^{q-1} e(r/q) \sum_{\substack{k \leq N \\ k \equiv r \pmod q}} \Lambda(k) \approx -\frac{N}{\phi(q)} \mu(q).$$

Right most side follows from the Siegel-Walfisz theorem. Thus when α is rational (or near a rational) and q is small, then $|S(\alpha)|$ is large. So to estimate (4.1) we split the integral into two parts one of which constitutes of the contribution of $S(\alpha)$ near rational numbers with small denominator.

4.1.1 Major and Minor Arcs

Discussion from the previous section motivates the following definitions.

Definition 4.2. *Let $P = (\log N)^B$, $Q = N(\log N)^{-B}$, where B is a positive constant. For $q \leq P$, and $1 \leq a \leq q$, with $(a, q) = 1$ we define $\mathfrak{M}(a, q)$ as*

$$\mathfrak{M}(a, q) = \left\{ \alpha \in \mathbb{R}/\mathbb{Z} : \left| \alpha - \frac{a}{q} \right| \leq \frac{1}{Q} \right\}.$$

We call this a major arc.

Define $\mathfrak{M}$ to be the union of all major arcs, i.e.,

$$\mathfrak{M} = \bigcup_{q \leq P} \bigcup_{\substack{1 \leq a \leq q \\ (a,q)=1}} \mathfrak{M}(a, q).$$

Remark 4.3. *The choice of major arcs depends on the problem being investigated.*

Note that for large N, the major arcs are disjoint. Indeed if we take $a/q \neq a'/q'$ we find

$$\left| \frac{a}{q} - \frac{a'}{q'} \right| \geq \frac{1}{qq'} \geq \frac{1}{P^2} > \frac{2}{Q},$$

for large N.
The strict inequality above shows that the compliment of $\mathfrak{M}$ in $[0, 1]$ is non empty. Therefore, we define $\mathfrak{m} = [0, 1] \backslash \mathfrak{M}$ and call it the minor arc.

We now split the integral (4.1) as

$$r(N) = \int_{\mathfrak{M}} S(\alpha)^3 e(-N\alpha) d\alpha + \int_{\mathfrak{m}} S(\alpha)^3 e(-N\alpha) d\alpha.$$

We shall now focus on estimating these two integrals.

4.2 Auxiliary lemmas

This section contains all the lemmas needed for the proof of theorem 4.1.

Definition 4.4. *For any character χ modulo q define*

$$\tau(\chi) = \sum_{m=1}^{q} \chi(m)e(m/q).$$

Lemma 4.5. *Let χ be a character modulo q, then*

$$\frac{1}{\phi(q)} \sum_{\chi} \tau(\overline{\chi})\chi(h) = \begin{cases} e(h/q) & \text{if } (h,q) = 1, \\ 0 & \text{if } (h,q) > 1. \end{cases}$$

Proof. If $(h,q) > 1$, $\chi(h) = 0$. So assume that $(h,q) = 1$. Then

$$\begin{aligned} \frac{1}{\phi(q)} \sum_{\chi} \tau(\overline{\chi})\chi(h) &= \frac{1}{\phi(q)} \sum_{\chi} \sum_{1 \le m \le q} \overline{\chi(m)}\chi(h)e(m/q) \\ &= \frac{1}{\phi(q)} \sum_{\chi} \sum_{1 \le a \le q} \overline{\chi(a)}e(ah/q) \\ &= \frac{1}{\phi(q)} \sum_{1 \le a \le q} \sum_{\chi} \overline{\chi(a)}e(ah/q) \\ &= \begin{cases} e(h/q) & \text{if } (h,q) = 1, \\ 0 & \text{if } (h,q) > 1. \end{cases} \end{aligned}$$

The last step follows from the orthogonality of characters modulo q. □

Lemma 4.6. *For any $q \in \mathbb{N}$ and χ_0 be the principle character modulo q we have $\tau(\chi_0) = \mu(q)$.*

Proof. It is well known that $\sum_{d|n} \mu(d) = \lfloor 1/n \rfloor$. Therefore we have

$$\begin{aligned} \tau(\chi_0) &= \sum_{\substack{1 \le k \le q \\ (k,q)=1}} e\left(\frac{k}{q}\right) \\ &= \sum_{1 \le k \le q} \sum_{d|(k,q)} \mu(d) e\left(\frac{k}{q}\right) \\ &= \sum_{d|q} \mu(d) \sum_{1 \le m \le q/d} e\left(\frac{md}{q}\right) \\ &= \mu(q). \end{aligned}$$

This completes the proof. □

Lemma 4.7. *For any $q \in \mathbb{N}$ and any character χ modulo q we have $\tau(\chi) \le \sqrt{q}$.*

Proof. See [1] chapter 9. We omit the proof here. □

Lemma 4.8. *(Weyl's Inequality) For and real number β and all integers $N_1 < N_2$,*

$$\sum_{n=N_1}^{N_2} e(n\beta) \ll \min(N_2 - N_1, \|\beta\|^{-1}).$$

Proof. By a trivial estimate it is clear that the right side is $\leq N_2 - N_1$. Since the sum on the right hand side is a geometric sum we have

$$\sum_{n=N_1}^{N_2} e(n\beta) = \frac{e(N_1\beta)(e((N_2 - N_1)\beta) - 1)}{e(\beta) - 1} \leq \frac{1}{|\sin(\pi\beta)|} = \frac{1}{\sin(\pi\{\beta\})}.$$

It suffices to note that for $\{\beta\} \leq 1/2$, $\sin(\pi\{\beta\}) \geq \{\beta\}$, and for $\{\beta\} > 1/2$, we get $\sin(\pi\{\beta\}) < 1 - \{\beta\}$. □

Lemma 4.9. *For any $x > 0$, we have*

$$\sum_{n>x} \frac{1}{\phi(n)^2} = O\left(\frac{1}{x}\right).$$

Proof. First note that for any $\varepsilon > 0$, we have $\phi(n) \gg n^{1-\varepsilon}$. This shows that the right hand side converges. Using this bound and the partial sum estimate using the zeta function we get the desired result. [ADD DETAILS!] □

4.3 Estimation of Sums of Prime Exponentials

We let

$$F(s) = \sum_{m \leq U} \Lambda(m) m^{-s}, \text{ and } G(s) = \sum_{d \leq V} \mu(d) d^{-s}.$$

Vaughan's identity gives

$$-\frac{\zeta'}{\zeta}(s) = F(s) - \zeta(s)F(s)G(s) - \zeta'(s)G(s) + \left(-\frac{\zeta'}{\zeta}(s) - F(s)\right)(1 - \zeta(s)G(s)), \quad (4.2)$$

valid for $\Re(s) > 1$. Comparing the nth Dirichlet's series coefficients, we get

$$\Lambda(n) = a_1(n) + a_2(n) + a_3(n) + a_4(n),$$

where

$$a_1(n) = \begin{cases} \Lambda(n) & \text{if } n \leq U, \\ 0 & \text{if } n > U; \end{cases}$$

$$a_2(n) = -\sum_{\substack{mdr=n \\ m \leq U \\ d \leq V}} \Lambda(m)\mu(d);$$

$$a_3(n) = \sum_{\substack{hd=n \\ d \leq V}} \mu(d) \log h;$$

and

$$a_4(n) = -\sum_{\substack{mk=n \\ m>U \\ k>1}} \Lambda(m) \left(\sum_{\substack{d|k \\ d\leq V}} \mu(d) \right).$$

We multiply throughout by $f(n)$ and sum, then

$$\sum_{n\leq N} f(n)\Lambda(n) = S_1 + S_2 + S_3 + S_4,$$

where

$$S_i = \sum_{n\leq N} f(n)a_i(n).$$

By the trivial bound on S_1, we get $S_1 \ll U$ for $|f(n)| \leq 1$. We treat the remaining S_i's individually.

Write S_2 in the form

$$S_2 = -\sum_{t\leq UV} \left(\sum_{\substack{md=t \\ m\leq U \\ d\leq V}} \mu(d)\Lambda(m) \right) \sum_{r\leq N/t} f(rt).$$

As $\sum_{m|t} \mu(d)\Lambda(m) = \log t \leq \log UV$. This gives

$$\sum_{\substack{md=t \\ m\leq U \\ d\leq V}} \mu(d)\Lambda(m) \ll \sum_{m|t} \Lambda(m) \ll \log UV.$$

Hence

$$S_2 \ll (\log UV) \sum_{t\leq UV} \left| \sum_{r\leq N/t} f(rt) \right|. \tag{4.3}$$

Taking out the μ factor we see that S_3 is of the same form as S_2, since

$$\begin{aligned} S_3 = \sum_{d\leq V} \mu(d) \sum_{h\leq N/d} f(dh)\log h &= \sum_{d\leq V} \mu(d) \sum_{h\leq N/d} f(dh) \int_1^h \frac{dz}{z} \\ &= \int_1^N \sum_{d\leq V} \mu(d) \sum_{h\leq N/d} f(dh)\mathbf{1}_{[1,h]}(z)\frac{dz}{z} \\ &= \int_1^N \sum_{d\leq V} \mu(d) \sum_{z\leq h\leq N/d} f(dh)\frac{dz}{z}. \end{aligned}$$

Therefore, we have

$$S_3 \ll (\log N) \sum_{d\leq V} \max_z \left| \sum_{z\leq h\leq N/d} f(dh) \right|. \tag{4.4}$$

The sum S_4 is more complicated. Note that

$$S_4 = \sum_{U<m\leq N/V} \Lambda(m) \sum_{V<k\leq N/m} \left(\sum_{\substack{d|k \\ d\leq V}} \mu(d) \right) f(mk)$$

since the sum vanishes for $1 < k \leq V$.

We partition the interval $(U, N/V]$ into dyadic intervals $(2^{r-1}U, 2^rU]$ where r is $\ll \log_2(N/2UV)$ and we let $\mathcal{S} = \left\{2^rU : 1 \leq r \leq \left\lfloor \log_2\left(\frac{N}{2UV}\right)\right\rfloor\right\} \cup \left\{\frac{N}{2V}\right\}$. This gives

$$S_4 \leq \sum_{M\in\mathcal{S}} \sum_{M<m\leq 2M} \Lambda(m) \sum_{V<k\leq N/m} \left(\sum_{\substack{d|k \\ d\leq V}} \mu(d) \right) f(mk).$$

Therefore, by applying Cauchy-Schwarz inequality on each term of the sum, we see that the Mth term is

$$\ll \left(\sum_{M<m\leq 2M} \Lambda(m)^2 \right)^{1/2} \left(\sum_{M<m\leq 2M} \left| \sum_{V<k\leq N/m} d(k) f(mk) \right|^2 \right)^{1/2},$$

where we have let

$$d(k) = \sum_{\substack{d|k \\ d\leq V}} \mu(d).$$

We expand the second

$$\sum_{M<m\leq 2M} \left| \sum_{V<k\leq N/m} d(k) f(mk) \right|^2 = \sum_{V<k\leq \frac{N}{M}} \sum_{V<j\leq \frac{N}{M}} d(k)\overline{d(j)} \sum_{\substack{M<m\leq 2M \\ m\leq N/k \\ m\leq M/j}} f(km)\overline{f(jm)}.$$

The trivial inequality $|d(k)\overline{d(j)}| \leq \frac{1}{2}(|d(k)|^2 + |d(j)|^2)$ shows that the square of the second term is

$$\ll \sum_{V<k\leq \frac{N}{M}} |d(k)|^2 \sum_{V<j\leq \frac{N}{M}} \left| \sum_{\substack{M<m\leq 2M \\ m\leq N/k \\ m\leq M/j}} f(km)\overline{f(jm)} \right|$$

$$\ll \left(\sum_{V<k\leq \frac{N}{M}} |d(k)|^2 \right) \left(\max_{V<j\leq \frac{N}{M}} \sum_{V<k\leq \frac{N}{M}} \left| \sum_{\substack{M<m\leq 2M \\ m\leq N/k \\ m\leq M/j}} f(km)\overline{f(jm)} \right| \right).$$

Finally we note that

$$\sum_{m\leq z}\Lambda(m)^2 \ll (\log z)\sum_{m\leq z}\Lambda(m) \ll z\log z,$$

and

$$\begin{aligned}\sum_{k\leq n} d(k)^2 &= \sum_{k\leq n}\frac{k\tau(k)^2}{k}\\ &\leq n\sum_{k\leq n}\frac{\tau(k)^2}{k}\\ &\leq n\prod_{p\leq n}\left(1+\frac{2^2}{p}+\frac{3^2}{p^3}+\cdots\right)\\ &\ll n\prod_{p\leq n}\left(1+\frac{1}{p}+\frac{1}{p^2}+\cdots\right)^4\\ &\ll n(\log n)^4.\end{aligned}$$

The last step follows from Merten's bound. Therefore, we have

$$S_4 \ll \sum_{M\in\mathcal{S}}\sqrt{M\log M}\sqrt{\frac{N}{M}}\left(\log\frac{N}{M}\right)^2\left(\max_{V<j\leq\frac{N}{M}}\sum_{V<k\leq\frac{N}{M}}\left|\sum_{\substack{M<m\leq 2M\\ m\leq N/k\\ m\leq M/j}} f(km)\overline{f(jm)}\right|\right)^{1/2}$$

$$\ll N^{1/2}(\log N)^{7/2}\max_{U\leq M\leq\frac{N}{V}}\left(\max_{V<j\leq\frac{N}{M}}\sum_{V<k\leq\frac{N}{M}}\left|\sum_{\substack{M<m\leq 2M\\ m\leq N/k\\ m\leq M/j}} f(km)\overline{f(jm)}\right|\right)^{1/2}. \tag{4.5}$$

Combining (4.3), (4.4), (4.5) and the trivial bound on S_1, we get

$$\sum_{n\leq N} f(n)\Lambda(n) \ll U + (\log N)\sum_{t\leq UV}\max_{z}\left|\sum_{z\leq r\leq N/t} f(rt)\right| \tag{4.6}$$

$$+ N^{1/2}(\log N)^{7/2}\max_{U\leq M\leq\frac{N}{V}}\left(\max_{V<j\leq\frac{N}{M}}\sum_{V<k\leq\frac{N}{M}}\left|\sum_{\substack{M<m\leq 2M\\ m\leq N/k\\ m\leq M/j}} f(km)\overline{f(jm)}\right|\right)^{1/2}$$

where $|f(n)|\leq 1$.

Remark 4.10. *A stronger result is true replacing the exponent* $7/2$ *by* $3/2$ *in the last term of (4.6). See [1] chapter 24.*

Theorem 4.11. *For a real number* α *such that* $|\alpha - a/q| \leq 1/q^2$*, where* $(a,q) = 1$ *we have*

$$S(\alpha) \ll (Nq^{-1/2} + N^{4/5} + N^{1/2}q^{1/2})(\log N)^4.$$

Proof. The bound is trivial when $q > N$, since then

$$S(\alpha) \ll N \log N \ll (Nq^{-1/2} + N^{4/5} + N^{1/2}q^{1/2})(\log N)^4.$$

We consider the other case. By Weyl's inequality we have

$$\sum_{t \leq UV} \max_{z} \left| \sum_{z \leq r \leq N/t} e(rt/\alpha) \right| \ll \sum_{t \leq UV} \min\left(\frac{N}{t}, \frac{1}{\|t\alpha\|}\right).$$

The latter expression is

$$\ll \left(\frac{N}{q} + UV + q\right)(\log 2qN)^2 \tag{4.7}$$

for $|\alpha - a/q| \leq 1/q^2$, which we shall prove later, but for now if we assume it. Then by (4.6) we get

$$S(\alpha) \ll U + \left(\frac{N}{q} + UV + q\right)(\log 2qN)^2 + N^{1/2}(\log N)^{7/2} \max_{U \leq M \leq N/V} \max_{V < j \leq N/M} \left(\sum_{V < k \leq N/M} \min\left(M, \frac{1}{\|(k-j)\alpha\|}\right)\right)^{1/2}.$$

Separating the case when $k = j$, the last term becomes

$$\ll N^{1/2}(\log N)^{7/2} \max_{U \leq M \leq N/V} \left(M + \sum_{1 \leq m \leq N/M} \min\left(\frac{N}{m}, \frac{1}{\|m\alpha\|}\right)\right)^{1/2},$$

which using (4.7), is

$$\ll N^{1/2}(\log N)^{7/2} \max_{U \leq M \leq N/V} \left(M + \frac{N}{M} + \frac{N}{q} + q\right)^{1/2} (\log qN)^{1/2}$$
$$\ll (NV^{-1/2} + NU^{-1/2} + Nq^{-1/2} + N^{1/2}q^{1/2})(\log qN)^4.$$

Therefore we have

$$S(\alpha) \ll (UV + q + NU^{-1/2} + NV^{-1/2} + Nq^{-1/2} + N^{1/2}q^{1/2})(\log qN)^4.$$

Since $q \leq N$, taking $U = V = N^{4/5}$ gives the desired result.

We now prove (4.7). Naturally we write $t = hq+r$ with $1 \le r \le q$, and put $\beta = \alpha - a/q$. Then

$$\sum_{t\le T} \min\left(\frac{N}{t}, \frac{1}{\|t\alpha\|}\right) \ll \sum_{0\le h\le T/q} \sum_{1\le r\le q} \min\left(\frac{N}{hq+r}, \left\|\frac{ra}{q} + hq\beta + r\beta\right\|^{-1}\right).$$

We break this sum as:

$$\sum_{0\le h\le T/q} \sum_{1\le r\le q/2} \min\left(\frac{N}{hq+r}, \left\|\frac{ra}{q} + hq\beta + r\beta\right\|^{-1}\right) + \sum_{0\le h\le T/q} \sum_{q/2<r\le q} \min\left(\frac{N}{hq+r}, \left\|\frac{ra}{q} + hq\beta + r\beta\right\|^{-1}\right)$$

To estimate the second sum we partition the interval $[0,1]$ into subintervals of length $1/(3q)$. If I is any interval of length $1/(3q)$ in $[0,1]$ then there are at most 3 values of $q/2 < r \le q$ for which

$$x = \frac{ra}{q} + hq\beta + r\beta \in I \pmod 1$$

which gives at most 6 values of $\|x\|$ in the inner sum. Also if for some r, $\|x\|^{-1} \le 1/q$, we take the other term from min. Therefore we see that the second term is

$$\ll \sum_{0\le h\le T/q} \left(\frac{N}{hq+\frac{q}{2}} + \sum_{1\le r\le q} \frac{3q}{r}\right)$$
$$\ll \frac{N}{q}\log\frac{T}{q} + T\log q.$$

By a similar argument we get that for $h > 0$, first term is bounded by this.

For the case when $h = 0$, we note that $\|ra/q\| - 1/(3q) \ll \|ra/q + r\beta\|$ and, as before, every interval of length $1/(3q)$ contains at most a constant number of $\|ra/q\|$. Therefore

$$\sum_{1\le r\le q/2} \frac{1}{\left\|\frac{ra}{q} + r\beta\right\|} \ll \sum_{1\le r\le q/2} \frac{1}{\left\|\frac{ra}{q}\right\| - \frac{1}{3q}} \ll \sum_{1\le r\le q/2} \frac{3q}{r} \ll q\log q.$$

On combining these estimates we get (4.7). □

4.4 Proof of Theorem 1

4.4.1 Contribution of Major Arcs

To estimate (4.1) we will first estimate the contribution of the major arcs. To this end we first determine the size of $S(\alpha)$ for $\alpha \in \mathfrak{M}(a,q)$. For $(a,q) = 1$, we let $\alpha = a/q + \beta$. Then, using lemma 4.5 we see

$$\sum_{\substack{k\le N\\(k,q)=1}} \Lambda(k)e(k\alpha) = \frac{1}{\phi(q)} \sum_{\chi} \tau(\overline{\chi})\chi(a) \sum_{k\le N} \chi(k)\Lambda(k)e(k\beta).$$

Also, note that the number of prime factors of q is $\ll \log q$. So if $p|q$ and l is such that $p^l \leq N$, then $\Lambda(p) + \cdots + \Lambda(p^l) = l \log p \ll \log N$. Therefore, we get

$$\sum_{\substack{k \leq N \\ (k,q)>1}} \Lambda(k) e(k\alpha) \ll \log q \log N \ll (\log N)^2,$$

so that

$$S(\alpha) = \frac{1}{\phi(q)} \sum_{\chi} \tau(\overline{\chi}) \chi(a) \sum_{k \leq N} \chi(k) \Lambda(k) e(k\beta) + O((\log N)^2). \tag{4.8}$$

Here, the inner sum is

$$\begin{aligned} \sum_{k \leq N} \chi(k) \Lambda(k) e(k\beta) &= \int_{1^-}^{N} e(u\beta) d(\psi(u, \chi)) \\ &= e(N\beta)\psi(N, \chi) - 2\pi i\beta \int_1^N e(u\beta)\psi(u, \chi) du \\ &\leq (1 + |\beta|N)|\psi(N, \chi)|. \end{aligned} \tag{4.9}$$

If $\chi \neq \chi_0$, then by Siegel-Walfisz theorem we see that $\psi(N, \chi) \ll N \exp(-c\sqrt{\log N})$, where c, a positive constant, only depends on B. Therefore, the inner sum is

$$\ll (1 + |\beta|N) N \exp(-c\sqrt{\log N}).$$

For the case when $\chi = \chi_0$, first note that

$$\psi(u) - \psi(u, \chi_0) = \sum_{\substack{k \leq u \\ (k,q)>1}} \Lambda(k) \ll \log q \log u.$$

Also, from the prime number theorem we know that $\psi(u) = u + O(u \exp(-c'\sqrt{\log u}))$. Thus, we obtain

$$\begin{aligned} \psi(u, \chi_0) &= \lfloor u \rfloor + O(u \exp(-c'\sqrt{\log u})) + O(\log q \log u) \\ &= \lfloor u \rfloor + R(u), \end{aligned}$$

where $R(u) = O(u \exp(-c\sqrt{\log u}))$ as we have taken $c = c'$ to keep things simple. Now we put $T(\beta) = \sum_{k \leq N} e(k\beta)$. Again by integration by part, we see that

$$T(\beta) = \int_{1^-}^{N} e(u\beta) d(\lfloor u \rfloor) = e(N\beta)N - 2\pi i\beta \int_1^N e(u\beta)\lfloor u \rfloor du.$$

By subtracting this from (4.9) we find that

$$\begin{aligned} \sum_{k \leq N} \chi_0(k) \Lambda(k) e(k\beta) &= T(\beta) + e(N\beta)R(N) - 2\pi i\beta \int_1^N e(u\beta)R(u) du \\ &= T(\beta) + O((1 + |\beta|N) N \exp(-c\sqrt{\log N})). \end{aligned}$$

Hence our sum has become

$$S(\alpha) = \frac{\tau(\chi_0)}{\phi(q)}(T(\beta) + O((1+|\beta|N|)N\exp(-c\sqrt{\log N}))) \\ + \frac{1}{\phi(q)}\sum_{\chi\neq\chi_0}\tau(\overline{\chi})\chi(a)O((1+|\beta|N)N\exp(-c\sqrt{\log N})).$$

To further simplify this expression, we use $\tau(\chi_0) = \mu(q)$ and $\tau(\chi) \leq \sqrt{q}$ from lemma 4.6 and 4.7.

These combined with that fact that $q \leq P$ and $|\beta| \leq 1/Q$ for $\alpha \in \mathfrak{M}(a,q)$ and that $(\log N)^B \ll N\exp(-c'\sqrt{\log N})$ for some $c' > 0$ we obtain

$$S(\alpha) = \frac{\mu(q)}{\phi(q)}T(\beta) + O(N\exp(-c_1\sqrt{\log N}))$$

for $\alpha \in \mathfrak{M}(a,q)$. Consequently

$$S(\alpha)^3 = \frac{\mu(q)}{\phi(q)^3}T(\beta)^3 + O(N^3\exp(-c_1\sqrt{\log N})),$$

and hence the contribution of $\mathfrak{M}(a,q)$ to the integral (4.1) is

$$\frac{\mu(q)}{\phi(q)^3}e\left(-\frac{aN}{q}\right)\int_{-1/Q}^{1/Q}T(\beta)^3e(-N\beta)d\beta + O(N^2\exp(-c_2\sqrt{\log N}))$$

where we have again used the fact that $(\log N)^B \ll N\exp(-c'\sqrt{\log N})$ for some $c' > 0$ to get the error term.
Summing over the various major arcs, we see that

$$\int_{\mathfrak{M}}S(\alpha)^3e(-N\alpha)d\alpha = \sum_{q\leq P}\frac{\mu(q)}{\phi(q)^3}c_q(N)\int_{-1/Q}^{1/Q}T(\beta)^3e(-N\beta)d\beta \\ + O(N^2\exp(-c_3\sqrt{\log N})), \tag{4.10}$$

where $c_3 > 0$ and $c_q(n)$ Ramanujan's sum,

$$c_q(n) = \sum_{\substack{1\leq a\leq q\\(a,q)=1}}e(an/q). \tag{4.11}$$

We now evaluate the first factor on the right hand side. By lemma 4.8 we see that

$$T(\beta) = \frac{e((N+1)\beta)-1}{e(\beta)-1} \ll \min(N, \|\beta\|^{-1}).$$

Hence

$$\int_{1/Q}^{1-1/Q}|T(\beta)^3|d\beta \ll Q^2 \ll N^2(\log N)^{-2B},$$

so that

$$\int_{-1/Q}^{1/Q} T(\beta)^3 e(-N\beta)d\beta = \int_0^1 T(\beta)^3 e(-N\beta)d\beta - \int_{1/Q}^{1-1/Q} T(\beta)^3 e(-N\beta)d\beta$$
$$= \int_0^1 T(\beta)^3 e(-N\beta)d\beta + O(N^2(\log N)^{2B}).$$

Note that

$$\int_0^1 T(\beta)^3 e(-N\beta)d\beta = \int_0^1 \sum_{k_1,k_2,k_3\leq N} e((k_1+k_2+k_3-N)\beta)d\beta$$
$$= \binom{N-1}{2} = \frac{1}{2}N^2 + O(N).$$

Hence

$$\int_{-1/Q}^{1/Q} T(\beta)^3 e(-N\beta)d\beta = \frac{1}{2}N^2 + O(N^2(\log N)^{-2B}). \tag{4.12}$$

To deal with the sum in (4.10) we first evaluate the Ramanujan's sum as follows

$$\sum_{1\leq a\leq q} e\left(\frac{an}{q}\right) = \sum_{d|q} \sum_{\substack{1\leq a\leq q \\ (a,q)=d}} e\left(\frac{an}{q}\right).$$

The sum on the left vanishes when $q \nmid n$, and is equal to q when $q|n$. Thus by Mobius inversion,

$$c_q(n) = \sum_{\substack{d|n \\ d|q}} d\mu\left(\frac{q}{d}\right). \tag{4.13}$$

Since the identity function and the mobius function are multiplicative, $c_q(n)$ is a multiplicative of q for any fixed n.

Let l be the highest power of prime p dividing n. From (4.11) we see that $c_{p^\beta}(n) = \phi(p^\beta)$, for $\beta \leq l$, $c_{p^{l+1}} = -p^l$, and using (4.13) $c_{p^\beta}(n) = 0$ for $\beta > l+1$. Thus

$$c_{p^\beta}(n) = \begin{cases} \phi(p^\beta) & \text{if } \beta \leq l, \\ -p^l & \text{if } \beta = l+1, \\ 0 & \text{if } \beta > l+1. \end{cases}$$

From the trivial estimate $|c_q(n)| \leq \phi(q)$ and by lemma 4.9, we see that

$$\sum_{q>P} \frac{\mu(q)}{\phi(q)^3} c_q(N) \ll \sum_{q>P} \frac{1}{\phi(q)^2} \ll (\log N)^{-B/2}$$

Since $c_q(n)$ is multiplicative for any fixed n, the sum in (4.10), when extended over all q, can be written as an absolutely convergent product,

$$\begin{aligned}\sum_{q=1}^{\infty}\frac{\mu(q)}{\phi(q)^3}c_q(N) &= \prod_p\left(1-\frac{c_p(N)}{(p-1)^3}\right)\\ &= \prod_{p|N}\left(1-\frac{1}{(p-1)^2}\right)\prod_{p\nmid N}\left(1+\frac{1}{(p-1)^3}\right)\\ &= \mathfrak{G}(N),\end{aligned}$$

so that

$$\sum_{q\le P}\frac{\mu(q)}{\phi(q)^3}c_q(N) = \mathfrak{G}(N)+O((\log N)^{-B/2}).$$

We combine this with (4.12) in (4.10) to get

$$\int_{\mathfrak{M}} S(\alpha)^3 e(-N\alpha)d\alpha = \frac{1}{2}\mathfrak{G}(N)N^2 + O(N^2(\log N)^{-B/2}). \tag{4.14}$$

Now we estimate the contribution of minor arcs.

4.4.2 Contribution of Minor Arcs

We note that

$$\begin{aligned}\left|\int_{\mathfrak{m}} S(\alpha)^3 e(-N\alpha)d\alpha\right| &\le \left(\max_{\mathfrak{m}}|S(\alpha)|\right)\int_{\mathfrak{m}}|S(\alpha)|^2\,d\alpha\\ &\le \left(\max_{\mathfrak{m}}|S(\alpha)|\right)\int_0^1|S(\alpha)|^2\,d\alpha.\end{aligned}$$

The last integral is

$$\sum_{k_1\le N}\sum_{k_2\le N}\Lambda(k_1)\Lambda(k_2)\int_0^1 e((k_1-k_2)\alpha)d\alpha = \sum_{1\le k\le N}\Lambda(k)^2 \ll N\log N.$$

For any real α Dirichlet's theorem on Diophantine approximation asserts that there is a rational number a/q such that $|\alpha - a/q| \le 1/qQ$, with $1\le q\le Q$, and $(a,q)=1$. Note that $q > P$, for if $q\le P$, then $\alpha\in\mathfrak{M}(a,q)$. Hence if $\alpha\in\mathfrak{m}$, then $P<q\le Q$, therefore by theorem 4.11

$$S(\alpha) \ll (Nq^{-1/2}+N^{4/5}+N^{1/2}q^{1/2})(\log N)^4 \ll Nq^{-1/2}(\log N)^4 \ll N(\log N)^{-(B/2)+4}.$$

Thus

$$\int_{\mathfrak{m}} S(\alpha)^3 e(-N\alpha)d\alpha \ll N^2(\log N)^{-(B/2)+5}.$$

This with (4.14) and $B = 2A+10$ gives the desired result. □

Bibliography

[1] H.L. Montgomery and H. Davenport. *Multiplicative Number Theory.* Graduate Texts in Mathematics. Springer New York, 2013.

[2] H.L. Montgomery and R.C. Vaughan. *Multiplicative Number Theory I: Classical Theory.* Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2007.

[3] M.B. Nathanson. *Additive Number Theory The Classical Bases.* Graduate Texts in Mathematics. Springer New York, 1996.

[4] E.M. Stein and R. Shakarchi. *Complex Analysis.* Princeton lectures in analysis. Princeton University Press, 2010.